B Differential Geometry and Tensor Calculations

B.1 Summary of Differential Manifolds

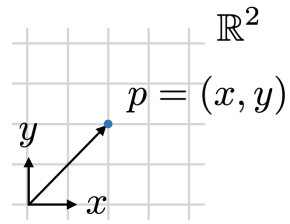


Fig. B.1: A point p in a Euclidean space $\mathbb{R}^2$.

We start the discussion in an Euclidean space $\mathbb{R}^2$. Given a point $p \in \mathbb{R}^2$, we can parameterize a point by two parameters x and y as coordinates, i.e., a point p is always written by $p = (x, y) = (x^1, x^2)$. For $\mathbb{R}^n$, a point p can be represented by the coordinates $(x^1, x^2, \cdots, x^n)$.

Hausdorff space. A topological space is called a *Hausdorff space* if the intersection of arbitrary two neighborhoods U and V of two points p and q, respectively, is empty, i.e., $U \cap V = \varnothing$.

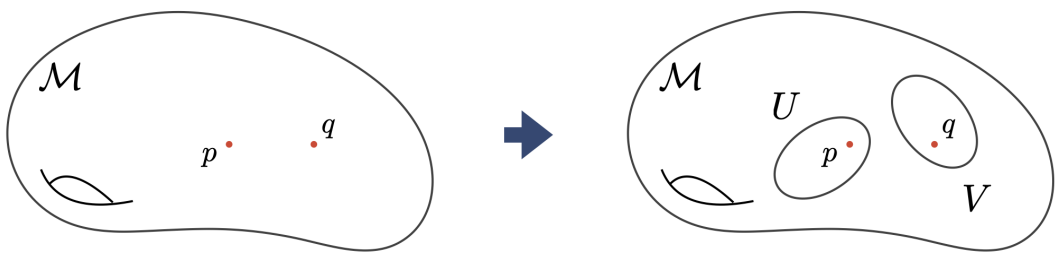


Fig. B.2: A Hausdorff space $\mathcal{M}$.

Manifold. Let $\mathcal{M}$ be a Hausdorff space. We say that $\mathcal{M}$ is called an n-dimensional *topological manifold* if the neighborhood of any point of $\mathcal{M}$ is isomorphic to an open set in Euclidean space $\mathbb{R}^n$.

Morphism. We will call a map the *morphism*, if it is *structure preserving*.

Local coordinates. Consider a point p in a neighborhood U of a manifold $\mathcal{M}$ with dimension n, we can define a local coordinate system (or a *coordinate chart* or a *coordinate patch* or a *parametrization*) of a neighborhood U of a point p and a *homeomorphism* (an *one-to-one* and *onto* map) φ as a pair (U, φ). The local coordinates of p is given by a map φ as a set of n-tuple real numbers in a neighborhood $D \subset \mathbb{R}^n$ as

$$\varphi : U \longrightarrow \mathbb{R}^n \tag{B.1a}$$

$$p \longmapsto \varphi(p) = x = (x^1, x^2, \cdots, x^n) \in \mathbb{R}^n. \tag{B.1b}$$

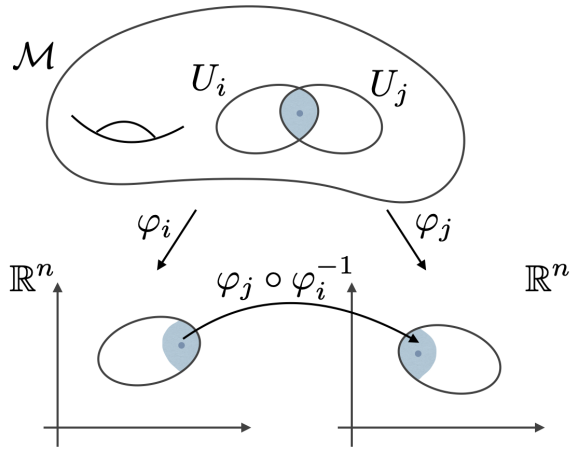


Fig. B.3: A manifold with coordinate charts.

Coordinate functions. The projection map u^μ sending the coordinate $x = (x^1, x^2, \cdots, x^n)$ of a point p to its μth-component is called the *coordinate function*, i.e., $x^\mu = u^\mu(x) = u^\mu \circ \varphi(p) \equiv \varphi^\mu(p)$.

Coordinate transformation. Consider two local coordinate systems (U_i, φ_i) and (U_j, φ_j) on $\mathcal{M}$. A point p in $U_i \cap U_j$ has two local coordinates $\varphi_i(p) = x_i$ and $\varphi_j(p) = x_j$. We define the *transition function*

$$\begin{cases} \varphi_{ji} \coloneqq \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j), & \text{(B.2a)} \\ \varphi_{ij} \coloneqq \varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \longrightarrow \varphi_i(U_i \cap U_j), & \text{(B.2b)} \end{cases}$$

and

$$\begin{cases} x_i \longmapsto x_j = \varphi_j \circ \varphi_i^{-1}(x_i) = \varphi_{ji}(x_i), & \text{(B.3a)} \\ x_j \longmapsto x_i = \varphi_i \circ \varphi_j^{-1}(x_j) = \varphi_{ij}(x_j). & \text{(B.3b)} \end{cases}$$

Differentiable manifold. The collection of all local coordinate systems is called an *atlas* of a manifold $\mathcal{M}$ defined by $\mathcal{A} = \{(U_i, \varphi_i)\}$. For any intersection of $U_i \cap U_j$, if φ_{ij} is r times continuously differentiable, then the atlas $\mathcal{A}$ is called the coordinate neighborhood system of class C^r. Therefore, if all the transformation functions are smooth (or differentiable), we say that the atlas $\mathcal{A}$ with the differentiability defines the *differential structure of class* C^r on the topological manifold $\mathcal{M}$.

Diffeomorphism. Consider another manifold $\mathcal{N}$, the diffeomorphism is a homeomorphism $f : \mathcal{M} \longrightarrow \mathcal{N}$ such that f and f^{-1} are both smooth. This is *globally* defined. *Locally*, given two local coordinate systems (U_i, φ_i) and (V_j, ψ_j) on $\mathcal{M}$ and $\mathcal{N}$, respectively, the induced coordinate transformation of the *local* diffeomorphism is read as

$$\begin{cases} x_i \longmapsto y_j = \psi_j \circ f \circ \varphi_i^{-1}(x_i) \coloneqq f_{ji}^{(yx)}(x_i), & \text{(B.4a)} \\ y_j \longmapsto x_i = \varphi_i \circ f^{-1} \circ \psi_j^{-1}(y_j) \coloneqq f_{ij}^{(xy)}(y_j). & \text{(B.4b)} \end{cases}$$

We note that $f_{ji}^{(yx)} = \psi_j \circ f \circ \varphi_i^{-1}$ and $f_{ij}^{(xy)} = \varphi_i \circ f^{-1} \circ \psi_j^{-1}$ are called the *expressions* of f in the local coordinates φ_i and ψ_j.

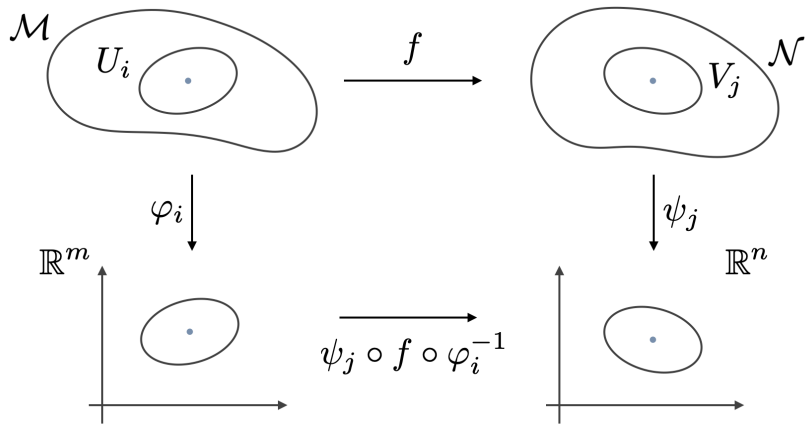


Fig. B.4: A map from manifold $\mathcal{M}$ to $\mathcal{N}$ with coordinate charts.

Automorphism. If $\mathcal{N} = \mathcal{M}$, i.e., f is a diffeomorphism from $\mathcal{M}$ to itself, $f : \mathcal{M} \longrightarrow \mathcal{M}$, we call f an *automorphism* of $\mathcal{M}$.

B.2 Tensor Fields

Scalar fields. A point x is characterized by a parameter t

Vector fields. Define a curve $\gamma : I_\varepsilon = (-\varepsilon, \varepsilon) \subset \mathbb{R} \longrightarrow \mathcal{M}$, which gives $\tau = \gamma(t)$ on a smooth manifold $\mathcal{M}$. Locally a point is given by $p = \gamma(0)$ on U and the corresponding parameterized coordinates should be

$$\varphi(\gamma(t)) = \varphi \circ \gamma(t) = x(t). \tag{B.5}$$

We assume that $f \in \mathfrak{F}(\mathcal{M}) : \mathcal{M} \longrightarrow \mathbb{R}$ is a function on $\mathcal{M}$ where $\mathfrak{F}(\mathcal{M})$ is the algebra of C^k. So the coordinate expression of f on U can be given by

$$f(p) = f\big(\varphi^{-1}(x)\big) = f \circ \varphi^{-1}(x) := F(x). \tag{B.6}$$

The maps including the coordinate chart from $\mathbb{R}$ to $\mathcal{M}$ to $\mathbb{R}$ is shown in Fig. B.5. Combine (B.5) and (B.6), a one-parameter varying function is

$$f\big(\gamma(t)\big) = f \circ \gamma(t) = \underbrace{f \circ \varphi^{-1}}_{F} \circ \underbrace{\varphi \circ \gamma}_{x}(t) = F \circ x(t) = F\big(x(t)\big). \tag{B.7}$$

A *tangent* vector or *velocity* vector filed at point $\gamma(t)$ with the local coordinates $\{x^\mu(t)\}$ in U can be obtained by the so-called derivative map

$$d\gamma : T_t\mathbb{R} \longrightarrow T_{\gamma(t)}\mathcal{M} \tag{B.8}$$

$$\frac{d}{dt} \longmapsto d\gamma\left(\frac{d}{dt}\right) \equiv \frac{d\gamma}{dt} = \dot{\gamma}(t) := X_{\gamma(t)}. \tag{B.9}$$

The tangent vector act on f would be

$$\frac{df\big(\gamma(0)\big)}{dt} = \frac{dF\big(x(0)\big)}{dt} = \frac{dx^\mu}{dt}(0) \cdot \frac{\partial F}{\partial x^\mu}\bigg|_{\gamma(0)}, \tag{B.10}$$

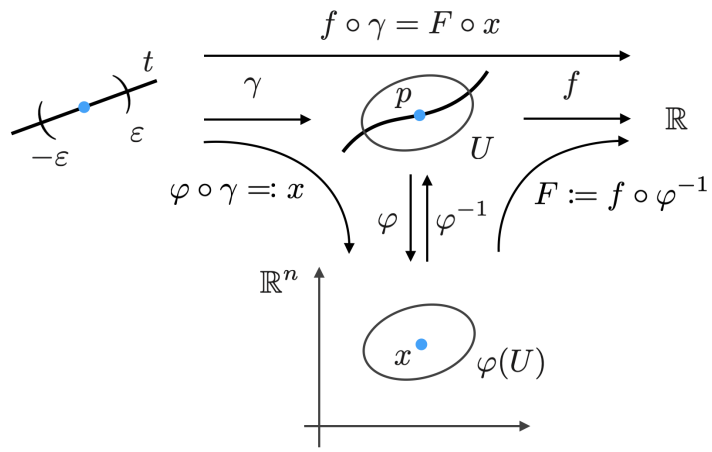


Fig. B.5: A map of a function f from $\mathbb{R}$ to $\mathbb{R}$.

where $\frac{\partial}{\partial x^\mu} \equiv \partial_\mu$. Hence, a vector field X at $\gamma(0) = p$ is defined as a map which sends a function $f \in \mathfrak{F}(\mathcal{M})$ to $\mathbb{R}$

$$X_p : \mathfrak{F}(\mathcal{M}) \longrightarrow \mathbb{R}, \tag{B.11}$$

and it has an expression of

$$X_p = X^\mu(p)\,(\partial_\mu)_p, \tag{B.12}$$

where the coefficients can be obtained by applying X to the coordinates x^μ, i.e.,

$$\boxed{X^\mu(p) = X_p(x^\mu)} = X^\nu(p)\frac{\partial x^\mu}{\partial x^\nu} = X^\nu(p)\delta^\mu_\nu. \tag{B.13}$$

It can obviously find that $X_p := X_{\gamma(0)}$ is a *derivation* of a smooth manifold $\mathcal{M}$ at point $\gamma(0) = p$, the set of all derivations is denoted by $D^r_p\mathcal{M}$ and

$$X_{\gamma(t)}(f) = X^\mu(t)\,\partial_\mu F|_{\gamma(t)}. \tag{B.14}$$

Therefore, by identification of $T_{\gamma(t)}\mathcal{M}$ and $D_{\gamma(t)}\mathcal{M}$, components of a tangent vector are given by a set of differential equations

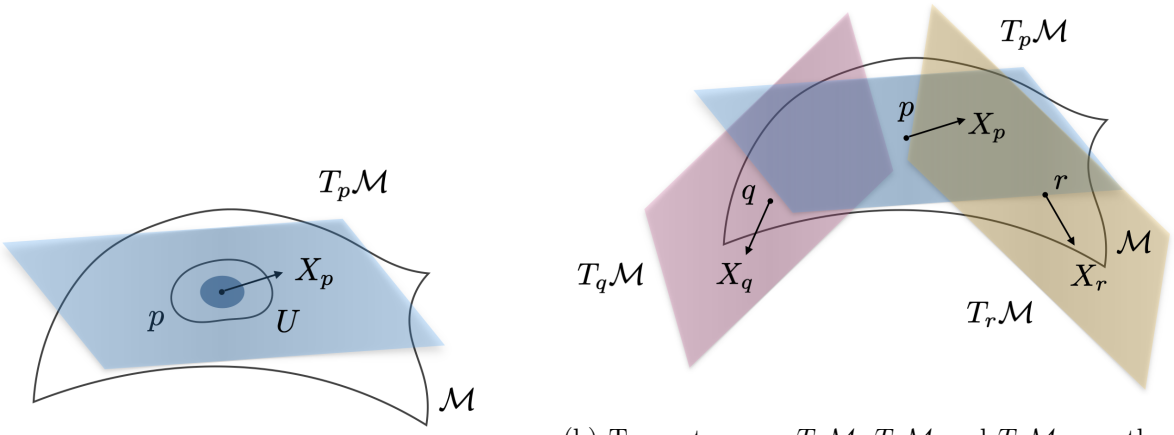
$$X^\mu(t) := \frac{dx^\mu}{dt}(t) \tag{B.15}$$

So the solution x^μ of (B.15) would be obtained

$$\boxed{x^\mu(t) = x^\mu(0) + \int_0^t dt'\,X^\mu(t')} = x^\mu(0) + t\,X^\mu_p + \mathcal{O}(t^2), \tag{B.16a}$$

where $X^\mu(t') = X^\mu(0) + t'\,dX^\mu(t')/dt' + \mathcal{O}(t'^2)$ and $X^\mu_p := X^\mu(0)$. Therefore, the tangent vector of the curve $\gamma(t)$ at p is a directional derivative along the direction of X

$$\dot\gamma(0) \equiv X_p = X^\mu_p(\partial_\mu)_p = X^\mu(0)(\partial_\mu)_{\gamma(0)}. \tag{B.17}$$



(a) A tangent space $T_p\mathcal{M}$ over a point p.

(b) Tangent spaces $T_p\mathcal{M}$, $T_q\mathcal{M}$ and $T_r\mathcal{M}$ over three point p, q and r respectively.

Fig. B.6: Tangent space.

The discussion above proves that every vector is a linear combination of $\{\partial_\mu\}$. This implies that a vector X represents a differential operator on $\mathcal{M}$. The basis vectors $\{\partial_\mu\}$ given by the local coordinates is called a *natural* basis. Moreover, we can define a so-called *tangent bundle* which is a collection of all tangent spaces

$$T\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p\mathcal{M}, \tag{B.18}$$

and the set of all differential vector fields on $\mathcal{M}$ is always denoted by $\mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$.

Consider the bracket operation of vectors X and $Y \in \mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$ acting on $f \in \mathfrak{F}(\mathcal{M})$, then we have

$$\begin{aligned}
[X,Y]f &= [X^\mu\partial_\mu, Y^\nu\partial_\nu]f \\
&= X^\mu(\partial_\mu Y^\nu)(\partial_\nu f) - Y^\nu(\partial_\nu X^\mu)(\partial_\mu f) \\
&= [X^\mu(\partial_\mu Y^\nu) - Y^\mu(\partial_\mu X^\nu)](\partial_\nu f) \\
&\equiv [X,Y]^\nu(\partial_\nu f),
\end{aligned} \tag{B.19}$$

which shows that $[X,Y]$ is a vector field. It implies that $\mathfrak{X}(\mathcal{M})$ is a Lie algebra. In addition we have the *Jacobi's identity*

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0. \tag{B.20}$$

Differential 1-forms. The map $f : \mathcal{M} \longrightarrow \mathbb{R}$ induces a derived map $(df)_p : T_p\mathcal{M} \longrightarrow T_{f(p)}\mathbb{R} \cong \mathbb{R}$ which is a linear function on a vector

$$(df)_p(aX + bY) = a(df)_p(X) + b(df)_p(Y) \quad \text{for } a, b \in \mathfrak{F}(\mathcal{M}) \text{ and } X, Y \in \mathfrak{X}(\mathcal{M}). \tag{B.21}$$

The function $(df)_p \in T_p^*\mathcal{M}$ is called a 1-*form* (a differential form of degree 1). Given a vector field $X \in \mathfrak{X}(\mathcal{M})$, we have

$$(df)_p(X_p) = \frac{df}{dt}(p) = X_p f \in \mathbb{R}. \tag{B.22}$$

Hence, we define a *cotangent bundle* as a set of all cotangent spaces

$$T^*\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p^*\mathcal{M}. \tag{B.23}$$

The set of all 1-form fields on $\mathcal{M}$ is then denoted by $A^1(\mathcal{M}) = \Gamma(T^*\mathcal{M})$.

Take $f = x^\mu$ and $X_p = (\partial_\nu)_p$. The component of local coordinates is $x^\mu : U \longrightarrow \mathbb{R}$, then $(dx^\mu)_p : T_pU \longrightarrow T_{x^\mu(p)}\mathbb{R} \cong \mathbb{R}$. Consequently we have property

$$(dx^\mu)_p\big((\partial_\nu)_p\big) = (\partial_\nu)_p(x^\mu) = \delta^\mu_\nu. \tag{B.24}$$

The equation above implies that $T^*\mathcal{M}$ is the dual space of $T\mathcal{M}$ and $\{dx^\mu\}$ forms a coordinate frame in $T^*\mathcal{M}$. Consequently, we can perform an inner product between $T\mathcal{M}$ and $T^*\mathcal{M}$:

$$\langle\ ,\ \rangle : T\mathcal{M} \times T^*\mathcal{M} \longrightarrow \mathbb{R}, \tag{B.25}$$

$$(X, \omega) \longmapsto \langle X, \omega\rangle. \tag{B.26}$$

Induced maps of vectors and differential 1-forms. Under a *coordinate transformation* between two charts of a point p (B.3), the transformations of natural basis of vectors and differential 1-forms can be understood by using the *chain rule*

$$\begin{cases} \dfrac{\partial}{\partial x_i^\mu} = \dfrac{\partial}{\partial x_j^\nu}\dfrac{\partial \varphi_{ji}^\nu}{\partial x_i^\mu} \equiv \dfrac{\partial}{\partial x_j^\nu}\dfrac{\partial x_j^\nu}{\partial x_i^\mu}, & \text{(B.27a)} \\[2ex] dx_i^\mu = \dfrac{\partial \varphi_{ij}^\mu}{\partial x_j^\nu}\, dx_j^\nu \equiv \dfrac{\partial x_i^\mu}{\partial x_j^\nu}\, dx_j^\nu. & \text{(B.27b)} \end{cases}$$

Given a diffeomorphism between two manifold $f : \mathcal{M} \longrightarrow \mathcal{N}$ such that $p \longmapsto q = f(p)$, a *pushforward* of vectors is induced

$$f_* : T_p\mathcal{M} \longrightarrow T_q\mathcal{N} = T_{f(p)}\mathcal{N}, \tag{B.28a}$$

$$X_p \longmapsto f_*X_p = (f_*X)_q = (f_*X)_{f(p)}. \tag{B.28b}$$

Let us choose coordinate systems in two manifolds (B.4), the pushforward of a basis vector from one to the other is

$$\left.\frac{\partial}{\partial x^\mu}\right|_p \longmapsto f_*\left(\left.\frac{\partial}{\partial x^\mu}\right|_p\right) = \left.\frac{\partial f_{yx}^\nu}{\partial x^\mu}\frac{\partial}{\partial y^\nu}\right|_{q=f(p)} = \left.\frac{\partial f_{yx}^\nu}{\partial x^\mu}\frac{\partial}{\partial y^\nu}\right|_{f(p)} \equiv \left.\frac{\partial y^\nu}{\partial x^\mu}\frac{\partial}{\partial y^\nu}\right|_{f(p)}. \tag{B.29}$$

Similarly, we can define a map with the *backward* direction with respect to the direction of f called a *pullback* of 1-forms induced by the map f

$$f^* : T_q^*\mathcal{N} \longrightarrow T_p^*\mathcal{M} = T_{f^{-1}(q)}^*\mathcal{M}, \tag{B.30a}$$

$$\omega_q \longmapsto f^*\omega_q = (f^*\omega)_p = (f^*\omega)_{f^{-1}(q)}. \tag{B.30b}$$

Therefore the pullback of a coordinate 1-form is

$$dy^\mu|_q \longmapsto f^*(dy^\mu|_q) = \left.\frac{\partial f_{yx}^\mu}{\partial x^\nu}\, dx^\nu\right|_{p=f^{-1}(q)} = \left.\frac{\partial f_{yx}^\mu}{\partial x^\nu}\, dx^\nu\right|_{f^{-1}(q)} \equiv \left.\frac{\partial y^\mu}{\partial x^\nu}\, dx^\nu\right|_{f^{-1}(q)}. \tag{B.31}$$

As discussions above, we compare properties of diffeomorphism, pushforward and pullback together as follows

$$\text{Diffeomorphism } f: \quad \mathcal{M} \text{ (left)} \longrightarrow \mathcal{N} \text{ (right)}, \tag{B.32a}$$

$$\text{Pushforward } f_*: \quad T_p\mathcal{M} \text{ (left)} \longrightarrow T_p\mathcal{N} \text{ (right)}, \tag{B.32b}$$

$$\text{Pullback } f^*: \quad T_p^*\mathcal{N} \text{ (right)} \longrightarrow T_p^*\mathcal{M} \text{ (left)}. \tag{B.32c}$$

It is apparent that the direction of pushforward f_* is the *same* as diffeomorphism f, and the direction of pullback f^*, however, is the *inversed*. That is why we called the objects with subscript the *covariant* fields and with superscript the *contravariant* fields.

Tensor fields. Suppose V is a n-dimensional vector space over $\mathfrak{F}$ and its dual space is V^*. We define (r,s)-type tensor fields V^r_s (or $V^{(r,s)}$) by the so-called tensor product

$$V^r_s : \underbrace{V^* \times \cdots \times V^*}_{r \text{ terms}} \times \underbrace{V \times \cdots \times V}_{s \text{ terms}} \longrightarrow \mathbb{R}. \tag{B.33}$$

Take e_μ as the basis of V and ϑ^μ as the basis of V^*. A tensor field $T \in V^r_s$ is then given by

$$\boxed{K = K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s} e_{\mu_1} \otimes \cdots \otimes e_{\mu_r} \otimes \vartheta^{\nu_1} \otimes \cdots \otimes \vartheta^{\nu_s},} \tag{B.34}$$

where

$$K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s} = K(\vartheta^{\mu_1}, \ldots, \vartheta^{\mu_r}, e_{\nu_1}, \ldots, e_{\nu_s}). \tag{B.35}$$

Obviously we have $V^0_0 = \mathfrak{F}$, $V^1_0 = V$ and $V^0_1 = V^*$.

Multilinear algebra. The spaces $\Lambda^r V$ and $\Lambda^r V^*$ being subspaces of V^r_0 and V^0_r are the set of all alternating contravariant and covariant tensor fields, respectively. Particularly, the space of all vectors is identified by $\Lambda^1 V = V = V^1_0$, while the space of all 1-forms is identified by $\Lambda^1 V^* = V^* = V^0_1$.

B.3 1-Parameter Group of Transformations

A 1-parameter group. Consider a mapping given by

$$\sigma : \mathbb{R} \times \mathcal{M} \longrightarrow \mathcal{M}, \tag{B.36}$$

$$(t,p) \longmapsto \sigma(t,p). \tag{B.37}$$

we can also have $\mathcal{M}$ by $\varphi_t(p) \equiv \gamma_p(t) := \sigma(t,p)$ where

$$\varphi_t : \mathcal{M} \ni p \longmapsto q = \varphi_t(p) \in \mathcal{M}, \tag{B.38a}$$

$$\gamma_p : \mathbb{R} \ni t \longmapsto \gamma_p(t) \in \mathcal{M}. \tag{B.38b}$$

For any $t \in \mathbb{R}$, the map φ_t is called the 1-parameter group of (differentiable) transformations of $\mathcal{M}$, which satisfies the following conditions:

- $\varphi_0 = \mathrm{id}_{\mathcal{M}}$, i.e., $\varphi_0(p) = \sigma(0,p) = p$;

- $\varphi_t \circ \varphi_s = \varphi_{t+s}$ for any $t, s \in \mathbb{R}$.

The second condition can be realized as follows:

$$(\varphi_t \circ \varphi_s)(p) = \gamma_{\varphi_s(p)}(t) = \gamma_p(t+s) = \sigma(t+s, p) = \varphi_{t+s}(p). \tag{B.39}$$

It can be shown that the conditions above implies $\varphi_{-t} = \varphi_t^{-1}$ by substituting $s = -t$. Therefore, $\varphi_t : \mathcal{M} \longrightarrow \mathcal{M}$ is a *diffeomorphism* from $\mathcal{M}$ to itself, which forms a diffeomorphism group denoted by $\mathrm{Diff}(\mathcal{M})$. This diffeomorphism also induces a map of tangent vectors $d\varphi_t : T\mathcal{M} \longrightarrow T\mathcal{M}$:

$$d\varphi_t(X_p) = X_q = X_{\varphi_t(p)}. \tag{B.40}$$

Let X be a vector field on $\mathcal{M}$, a parameterized curve γ_p called the *orbit* of φ_t through $p \in \mathcal{M}$. By writting as $\varphi_t(p)$, it can also be called an *integral curve* (or a *flow*) of X through a point p at $t = 0$. Therefore, we say that the transformation induces a *complete* vector field X on $\mathcal{M}$. Conversely, any $X \in \mathfrak{X}(\mathcal{M})$ generates a 1-parameter group of transformations of $\mathcal{M}$, $\varphi_t \in \mathrm{Diff}(\mathcal{M}) : \mathcal{M} \longrightarrow \mathcal{M}$.

A local version of transformation can be defined in a same way, if t is defined in a neighborhood of 0 and p is in an open set U of $\mathcal{M}$. The induced vector field is

$$X(t,p) \equiv X_{\gamma_p(t)} = d\gamma_p\left(\frac{d}{dt}\right) = \frac{d\gamma_p(t)}{dt} = \frac{d}{dt}\sigma(t,p). \tag{B.41}$$

Let x be a local coordinates of U, then the vector component in U should be

$$X^\mu(t,p) = X_{\gamma_p(t)}(x^\mu) = d\gamma_p\left(\frac{d}{dt}\right)(x^\mu) = \frac{d}{dt}\big(x^\mu \circ \gamma_p(t)\big) = \frac{d}{dt}x^\mu\big(\sigma(t,p)\big). \tag{B.42}$$

Similarly, any $X \in \mathfrak{X}(U)$ generates a local 1-parameter group of *local* transformations of U, $\varphi_t \in \mathrm{Diff}(U) : U \longrightarrow U$. By Taylor expansion, we have a local diffeomorphism from a point p to $\sigma(t,p) \equiv q$:

$$q = \sigma(t,p) = \sigma(0,p) + t\,X(0,p) + \mathcal{O}(t^2) = p + t\,X_p + \mathcal{O}(t^2), \tag{B.43}$$

which can be expressed in the local coordinate system in linear term

$$x^\mu(t,p) = x^\mu(0,p) + t\,X^\mu(0,p) \quad \text{or} \quad x^\mu(q) = x^\mu(p) + t\,X^\mu_p, \tag{B.44}$$

where we have defined $x^\mu(t,p) := x^\mu\big(\sigma(t,p)\big) \equiv x^\mu(q)$.

Exponential map. The group...

B.4 Lie Derivative

Infinitesimal transformation. Infinitesimally we have the exponential map from the Lie algebra to its Lie group

$$\varphi_t = \exp(t\,X) = 1 + t\,X + \frac{1}{2!}t^2\,X^2 + \cdots\,. \tag{B.45}$$

This infinitesimal transformation p to

$$\varphi_t^\mu(p) \approx x^\mu + t\,X(x^\mu) = x^\mu + t\,X^\nu\,\underbrace{\partial_\nu x^\mu}_{\delta^\mu_\nu} = x^\mu + t\,\xi^\mu \tag{B.46}$$

Lie derivative for a scalar. The Lie derivative for a scalar is defined by

$$\boxed{\pounds_X f = \lim_{t\to 0} \frac{(\varphi_t)^* f - f}{t} = \frac{d}{dt}(\varphi_t)^* f\bigg|_{t=0}.} \tag{B.47}$$

Act it on a point p, we get

$$(\pounds_X f)(p) = \lim_{t\to 0} \frac{((\varphi_t)^* f)(p) - f(p)}{t} = \lim_{t\to 0} \frac{(f\circ\varphi_t)(p) - f(p)}{t} \tag{B.48}$$

where

$$(f\circ\varphi_t)(p) = \underbrace{f(\varphi_t(p))}_{f(q)} = f(p + tX_p) = f(p) + tX_p(f). \tag{B.49}$$

So the equation above becomes

$$(\pounds_X f)(p) = \lim_{t\to 0} \frac{f(p) + tX_p(f) - f(p)}{t} = X_p(f) = Xf(p) \tag{B.50}$$

Finally the Lie derivative for a scalar is

$$\boxed{\pounds_X f = Xf.} \tag{B.51}$$

Lie derivative for a vector. The Lie derivative for a vector is defined by

$$\boxed{\begin{aligned}\pounds_X Y &= \lim_{t\to 0} \frac{Y - (\varphi_t)_* Y}{t} \\ &= \lim_{t\to 0} (\varphi_t)_* \frac{(\varphi_t^{-1})_* Y - Y}{t} = \lim_{t\to 0} \frac{(\varphi_{-t})_* Y - Y}{t} = \frac{d}{dt}(\varphi_{-t})_* Y\bigg|_{t=0},\end{aligned}} \tag{B.52}$$

where we have use the fact of $(\varphi_0)_* = \mathrm{id}$. Act this on a function f more precisely leads to

$$(\pounds_X Y)_q f = \lim_{t\to 0} \frac{Y_q f - ((\varphi_t)_* Y)_q f}{t} \tag{B.53}$$

We note that the second term can be realized by

$$((\varphi_t)_* Y)_q f = (Y(f\circ\varphi_t))(q(t)) = (Yf)(q(t)) + t\cdot Y(Xf)(q(t)) \tag{B.54}$$

where we have used $q(t) \coloneqq \varphi_t^{-1}(q) \equiv p$ and

$$f\circ\varphi_t = f + t\cdot(Xf). \tag{B.55}$$

Hence, we have

$$(\pounds_X Y)_q f = \lim_{t\to 0} \frac{(Yf)(q) - \big[(Yf)(q(t)) + t\cdot Y(Xf)(q(t))\big]}{t}$$

$$
\begin{aligned}
&= \underbrace{\lim_{t\to 0} \frac{(Yf)(q) - (Yf)(q(t))}{t}}_{\text{Let } g = Yf,\ \lim_{t\to 0} \frac{g(q)-g(q(t))}{t} = X_{q(0)}(g) = X_q(g)} - Y(Xf)(q(0)) \\
&= X(Yf)(q) - Y(Xf)(q) \\
&= [X, Y]_q f. \qquad \text{(B.56)}
\end{aligned}
$$

As a result, the Lie derivative for a vector is

$$
\boxed{\pounds_X Y = [X, Y].} \qquad \text{(B.57)}
$$

Lie derivative for a 1-form.

$$
\boxed{\pounds_X \omega = \lim_{t\to 0} \frac{(\varphi_t)^* \omega - \omega}{t}} \qquad \text{(B.58)}
$$

Lie derivative for a tensor. For tensor fields, we have a operation denoted by $(\widetilde{\varphi}_t)^*$ which combines operations of $(\varphi_{-t})_*$ for vectors and $(\varphi_t)^*$ for scalars and 1-forms

$$
\boxed{\pounds_X T = \lim_{t\to 0} \frac{(\widetilde{\varphi}_t)^* T - T}{t}} \qquad \text{(B.59)}
$$

Lie differential of a tensor. Finally we can define the *Lie differential* of a tensor T

$$
\delta_L T = (\widetilde{\varphi}_t)^* T - T \quad \text{or} \quad T - (\widetilde{\varphi}_t)_* T, \qquad \text{(B.60)}
$$

then we can conclude that

$$
\boxed{\delta_L T = \delta t\, \pounds_X T \quad \text{or} \quad \pounds_X T = \frac{\delta_L T}{\delta t}} \qquad \text{(B.61)}
$$

The details of calculation are associated with the *infinitesimal point transformation* which can be found in Appendix C.

B.5 Lie Groups, Lie Algebras and Their Representations

Lie group. Consider a group G which is a *differetiable* manifold, then G is a Lie group if its inverse map

$$
\tau : G \longrightarrow G, \qquad \text{(B.62a)}
$$

$$
g \longmapsto g^{-1} = \tau(g) \qquad \text{(B.62b)}
$$

and multiplication map

$$
\phi : G \times G \longrightarrow G, \qquad \text{(B.63a)}
$$

$$
(g, h) \longmapsto \phi(g, h) = gh \qquad \text{(B.63b)}
$$

are both *smooth*.

There are two operations of a Lie group. Given $g \in G$, we have *left translation* (resp. *right translation*)

$$L_g : G \longrightarrow G, \tag{B.64a}$$

$$a \longmapsto L_g(a) = ga, \tag{B.64b}$$

reps.

$$R_g : G \longrightarrow G, \tag{B.65a}$$

$$a \longmapsto R_g(a) = ag. \tag{B.65b}$$

Lie algebra. For both left (reps. right) transition, we obtain tangent maps

$$L_{g*a} : T_aG \longrightarrow T_aG, \tag{B.66a}$$

$$X_a \longmapsto L_{g*a}(X_a), \tag{B.66b}$$

reps.

$$R_{g*a} : T_aG \longrightarrow T_aG, \tag{B.67a}$$

$$X_a \longmapsto R_{g*a}(X_a), \tag{B.67b}$$

Particularly, we define a *left* (resp. *right*) *invariant vector fields*

$$L_{g*}(\widetilde{X}) = \widetilde{X}, \quad \text{resp.} \quad R_{g*}(\widetilde{X}) = \widetilde{X}. \tag{B.68}$$

More precise, it should be read as

$$L_{g*a}(\widetilde{X}_a) = \widetilde{X}_{ga}, \quad \text{resp.} \quad R_{g*a}(\widetilde{X}_a) = \widetilde{X}_{ag}. \tag{B.69}$$

The set of all left (resp. right) invariant vector fields are denoted by $\mathfrak{X}^L(G)$ (resp. $\mathfrak{X}^R(G)$). For $a = e \in G$ with the identification $X \coloneqq \widetilde{X}_e$, we obtain equation under the left (reps. right) translation

$$L_{g*e}(X) = \widetilde{X}_{ge} = \widetilde{X}_g, \quad \text{resp.} \quad R_{g*e}(X) = \widetilde{X}_{eg} = \widetilde{X}_g, \tag{B.70}$$

which implies that $X \in T_eG$ generates a left (reps. right) invariant vector field $\widetilde{X}_g$. We can show that the commutator

$$L_{g*}[\widetilde{X}, \widetilde{Y}] = [L_{g*}\widetilde{X}, L_{g*}\widetilde{Y}] = [\widetilde{X}, \widetilde{Y}], \quad \text{resp.} \quad R_{g*}[\widetilde{X}, \widetilde{Y}] = [R_{g*}\widetilde{X}, R_{g*}\widetilde{Y}] = [\widetilde{X}, \widetilde{Y}]. \tag{B.71}$$

is also a left (reps. right) invariant vector field in $\mathfrak{X}^{L/R}(G) = \Gamma(TG)$ with $TG = \bigcup_{g \in G} T_gG$.

The set of all left invariant vector fields $\mathfrak{X}^L(G)$ togather with the bracket multiplication product $[\widetilde{X}, \widetilde{Y}] = \widetilde{X}\widetilde{Y} - \widetilde{Y}\widetilde{X}$ defines a *Lie algebra* $\mathfrak{g}$ of a Lie group G, i.e.,

$$\boxed{\text{Lie algebra } \mathfrak{g} = \text{Invariant vector space } \mathfrak{X} + \text{Lie bracket } [\bullet, \bullet].} \tag{B.72}$$

It can be proved that the commutator

$$[X, Y] \coloneqq [\widetilde{X}, \widetilde{Y}]_e \tag{B.73}$$

is also a left invariant vector field

$$L_{g*}[X, Y] = L_{g*}[\widetilde{X}, \widetilde{Y}]_e = [\widetilde{X}, \widetilde{Y}]_g. \tag{B.74}$$

Therefore, we says that T_eG is isomorphic to the Lie algebra $\mathfrak{g}$, and T_eG is also a Lie algebra of G.

Structure constant. Let $\{X_i\}$ be a basis in $\mathfrak{g}$ of a Lie group G, the commutator of two bases satisfies

$$[X_i, X_j] = X_k \, c^k{}_{ij}, \tag{B.75}$$

where $c^k{}_{ij} = -c^k{}_{ji}$ is called the *structure constants* of G. If $\{Y_i\}$ is the basis of the space of the *right* invariant fields, then

$$[Y_i, Y_j] = Y_k \, \bar{c}^k{}_{ij}. \tag{B.76}$$

It can be checked that $\bar{c}^k{}_{ij} = -c^k{}_{ij}$.

For example $G = \mathrm{GL}(n; \mathbb{R})$, the local coordinate is x^{ij}, which gives a natural frame

$$\{E_{ij}\} \coloneqq \left\{ \frac{\partial}{\partial x^{ij}} \right\} \tag{B.77}$$

of a Lie algebra $\mathfrak{gl}(n, \mathbb{R})$ and satisfies

$$[E_{ij}, E_{kl}] = E_{rs} \, c^{rs}{}_{ij\,kl}. \tag{B.78}$$

Exponential map. An exponential map is a map from $\mathfrak{g}$ to G:

$$\exp : \mathfrak{g} \longrightarrow G, \tag{B.79}$$

$$X \longmapsto \exp X = e^X. \tag{B.80}$$

Representation. A *representation* is always indicates as a *linear* transformation from the space to itself. For *nonlinear* case, some one can emphasize it as a *realization*.

Consider an n-dimensional manifold $\mathcal{M}$, a representation of order n of the Lie group G is define by a map

$$\rho : G \longrightarrow \mathrm{GL}(G), \tag{B.81a}$$

$$g \longmapsto \rho(g) \equiv \rho_g, \tag{B.81b}$$

$$\rho_g : \mathcal{M} \longrightarrow \mathcal{M}, \tag{B.81c}$$

$$p \longmapsto \rho_g(p) = gp. \tag{B.81d}$$

It is obvious that the representation is a *left action* of group G on $\mathcal{M}$, i.e.,

$$\phi : G \times \mathcal{M} \longrightarrow \mathcal{M}, \tag{B.82a}$$

$$(g, p) \longmapsto \phi(g, p) \coloneqq L_g(p) = gp = \rho_g(p). \tag{B.82b}$$

One can also have a map

$$\sigma : \mathcal{M} \times G \longrightarrow \mathcal{M}, \tag{B.83a}$$

$$(p, g) \longmapsto \sigma(p, g) \coloneqq R_g(p) = pg. \tag{B.83b}$$

This defines a *right action* of G on $\mathcal{M}$, which is also a representation of G on $\mathcal{M}$.

Group action.

Effective. A group action of G on $\mathcal{M}$ is *effective* if $gp = p$ implies $g = e$.
Free. A group action of G on $\mathcal{M}$ is *free* if $gp = p$ implies $g = e$.
Transitive. A group action of G on $\mathcal{M}$ is *transitive* if

We now consider the case of n-dimensional Lie group G acting on itself. For any $g \in G$, an *inner automorphism* of G is defined by a conjugation by g

$$C_g : G \longrightarrow G, \tag{B.84a}$$

$$h \longmapsto \boxed{C_g(h) = ghg^{-1}.} \tag{B.84b}$$

The inner automorphisms $C_g \equiv C(g)$ forms a subgoup of an *automorphism* group $\mathrm{Aut}(G)$, which is called the inner automorphism group $\mathrm{Inn}(G) \equiv C(G)$:

$$C_g = L_g \circ R_{g^{-1}} = R_{g^{-1}} \circ L_g. \tag{B.85}$$

Hence, for a vector field $X \in \mathfrak{g}$, C_g with $g \in G$ induces a homomorphism $\mathrm{Ad}_g := C_{g*}$ from $\mathfrak{g}$ to itself, i.e., an inner automorphism of $\mathfrak{g}$, $\mathrm{Inn}(\mathfrak{g})$:

$$\mathrm{Ad}_g : \mathfrak{g} \longrightarrow \mathfrak{g}, \tag{B.86a}$$

$$X \longmapsto \boxed{\mathrm{Ad}_g(X) = R_{g^{-1}*} \circ L_{g*}(X) = R_{g^{-1}*}(X),} \tag{B.86b}$$

which implies the map Ad is a group homomorphism

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(n;\mathbb{R}) = \mathrm{GL}(\mathfrak{g}), \tag{B.87a}$$

$$g \longmapsto \mathrm{Ad}(g) := \mathrm{Ad}_g \tag{B.87b}$$

from G to $\mathrm{GL}(\mathfrak{g})$ called the *adjoint representation of the Lie group G on $\mathfrak{g}$*.

Furthermore, the adjoint representation of G induces a differential map $\mathrm{ad} := \mathrm{Ad}_* = d(\mathrm{Ad})$ from $\mathfrak{g}$ to $\mathfrak{gl}(\mathfrak{g}) = \mathrm{End}(\mathfrak{g})$ called the *adjoint representation of the Lie algebra $\mathfrak{g}$ of the Lie group G on $\mathfrak{g}$*:

$$\mathrm{ad} : \mathfrak{g} \longrightarrow \mathrm{End}(\mathfrak{g}), \tag{B.88a}$$

$$X \longmapsto \mathrm{ad}(X) := \mathrm{ad}_X\,. \tag{B.88b}$$

Hence, given a vector $Y \in \mathfrak{g}$, we have a map

$$\mathrm{ad}_X : \mathfrak{g} \longrightarrow \mathfrak{g}, \tag{B.89a}$$

$$Y \longmapsto \boxed{\mathrm{ad}_X(Y) = [X, Y] = XY - YX.} \tag{B.89b}$$

Take $X = T_i$ and $Y = T_j$ to be two bases in $\mathfrak{g}$, then we have the transformation of T_b:

$$\mathrm{ad}_{T_i}(T_j) = [T_i, T_j] = T_iT_j - T_jT_i \tag{B.90a}$$

$$= T_k\, c^k{}_{ij} \equiv T_k\ \mathrm{ad}(T_i)^k{}_j, \tag{B.90b}$$

which implies the adjoint representation of T_i can be identified by the structure constant ${c^k}_{ij}$

$$\boxed{\mathrm{ad}(T_i)^k{}_j := {c^k}_{ij}.} \tag{B.91}$$

The Killing form/metric (or Cartan-Killing form/metric) is a bilinear form $K_{\mathfrak{g}}$ on $\mathfrak{g}$ defined by

$$K_{\mathfrak{g}}(X,Y) = \mathrm{Tr}[\mathrm{ad}(X)\,\mathrm{ad}(Y)]. \tag{B.92}$$

Let $X = T_i$ and $Y = T_j$, the Killing form should be

$$\boxed{g_{ij} = K_{\mathfrak{g}}(T_i,T_j) = \mathrm{Tr}[\mathrm{ad}(T_i)\,\mathrm{ad}(T_j)] = \mathrm{ad}(T_i)^k{}_\ell\ \mathrm{ad}(T_j)^\ell{}_k = {c^k}_{i\ell}\,{c^\ell}_{jk}.} \tag{B.93}$$

Exponential map. Given $A \in \mathfrak{g}$, A can be used to generate G by one parameter subgroup group of G

$$\boxed{a(t) = \exp tA,} \tag{B.94}$$

which is called the *exponential map* and satisfies the differential equation

$$\frac{da(t)}{dt} = a(t)\,A \quad \text{with} \quad a(0) = I. \tag{B.95}$$

We can write $a(t)$ as an expansion of

$$a(t) = I + tA + \frac{t^2A^2}{2!} + \frac{t^3A^3}{3!} + \mathcal{O}(t^4). \tag{B.96}$$

For a principal bundle P, the structure group act on P on the right. For any $u \in P$, the orbit is $u\,a(t) = u\exp tA$ at $t = 0$. There is a *fundamental vector field* $A^*_u \in T_uP$ corresponding to $A \in \mathfrak{g}$. The action of G gives a map $P \times G \ni (u,a) \longmapsto ua \in P$ which induces a map $TP \times TG \ni (u,A) \longmapsto uA \in TP$ due to $u \in P \subset TP$ and $A \in \mathfrak{g} \subset TG$. So we can define the fundamental vector field by

$$\boxed{A^*_u = uA.} \tag{B.97}$$

Maurer-Cartan form. Given $g \in G$, a differential form ϑ is called left invariant if $L^*_g\vartheta = \vartheta$. We define the space of all the left invariant 1-form dual to the Lie algebra $\mathfrak{g}$ by $\mathfrak{g}^*$. The differential 2-form

$$d\vartheta = -\frac{1}{2}\,[\vartheta,\vartheta]. \tag{B.98}$$

Given two vector $X, Y \in \mathfrak{g}$, the action of a two form $d\omega$ is

$$\boxed{d\omega(X,Y) = \frac{1}{2}\big[X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y])\big].} \tag{B.99}$$

Let $X = T_i$, $Y = T_j$ and $\omega = \vartheta^k$, we obtain

$$d\vartheta^k(T_i,T_j) = \frac{1}{2}\big[\underbrace{T_i(\vartheta^k(T_j))}_{0} - \underbrace{T_j(\vartheta^k(T_i))}_{0} - \vartheta^k([T_i,T_j])\big] = -\frac{1}{2}\,{c^\ell}_{ij}\,\vartheta^k(T_\ell) = -\frac{1}{2}\,{c^k}_{ij} \tag{B.100}$$

As a result,

$$\boxed{d\vartheta^k = -\frac{1}{2}\,{c^k}_{ij}\,\vartheta^i \wedge \vartheta^j.} \tag{B.101}$$

B.6 Fibre Bundles

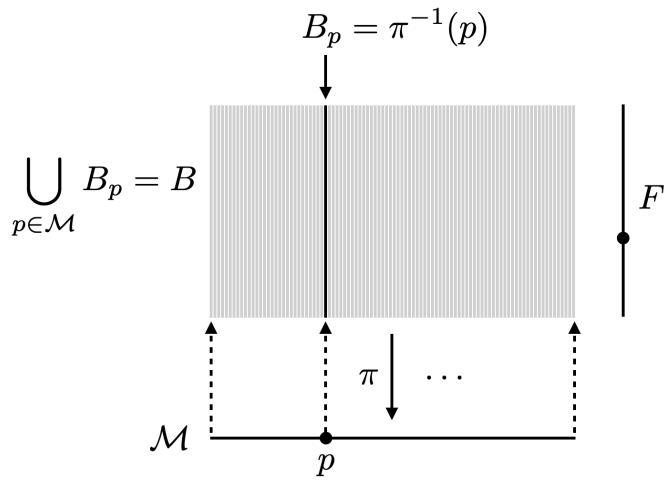


Fig. B.7: A fibre bundle.

Fibre bundle. A bundle $\mathcal{B}$ consists of three object: a manifold $\mathcal{M}$ is the base space; a bundle space denoted by B is called the *total space* over $\mathcal{M}$; a projection

$$\pi : B \longrightarrow \mathcal{M}, \tag{B.102a}$$
$$u \longmapsto \pi(u). \tag{B.102b}$$

We always write the bundle as a triple $\mathcal{B} = (B, \mathcal{M}, \pi)$. For any $p \in \mathcal{M}$, we have a fibre denoted by $B_p := \pi^{-1}(p)$ over p. A space of a standard (or a typical) fibre is denoted by F, which is homeomorphic to B_p. Now we make the bundle as a differetial manifold, i.e., there is a *differentiable structure* on B.[†16] For any point p, there is a aneighborhood $U_i \subset \mathcal{M}$, such that B can be *locally* treated as a space isomorphic to the direct product of U_i and F. For any $u \in \pi^{-1}(U_i)$, we can locally have another projection

$$\psi_i : \pi^{-1}(U_i) \longrightarrow F, \tag{B.103a}$$
$$u \longmapsto \psi_i(u). \tag{B.103b}$$

Hence, there is a diffeomorphism

$$\phi_i : \pi^{-1}(U_i) \longrightarrow U_i \times F, \tag{B.104a}$$
$$u \longmapsto \phi_i(u) = \big(\pi(u), \psi_i(u)\big), \tag{B.104b}$$

called a *local trivialization* of the total space B. For a family of neighborhood $\{U_i\}$ of $p \in \mathcal{M}$, transition functions are defined by

$$\phi_i(u) \cdot \phi_j^{-1}(u) =: g_{ij} : U_i \cap U_j \longrightarrow G, \tag{B.105}$$

[†16] A topology is introduced on B such that it becomes a Hausdorff space, and then defines a differentiable structure to make it a smooth manifold.

where G is called the *structure group* of B. For any $p \in U_i \cap U_j$, there is $g_{ij}(p) \in G$ acting a the element

$$f_i(p) \coloneqq \psi_i(u) = \psi_i \circ \pi^{-1}(p) \tag{B.106}$$

of the fibre F on the left:

$$g_{ij}(p) : F \longrightarrow F, \tag{B.107a}$$

$$f_j \longmapsto \boxed{f_i = g_{ij}(p) \cdot f_j.} \tag{B.107b}$$

We remark that transition functions have the following properties

$$g_{ii}(p) = e \quad \text{for} \quad p \in U_i, \tag{B.108a}$$

$$g_{ji}(p) = g_{ij}^{-1}(p) \quad \text{for} \quad p \in U_i \cap U_j, \tag{B.108b}$$

$$g_{ij}(p) \circ g_{jk}(p) = g_{ik}(p) \quad \text{for} \quad p \in U_i \cap U_j \cap U_k. \tag{B.108c}$$

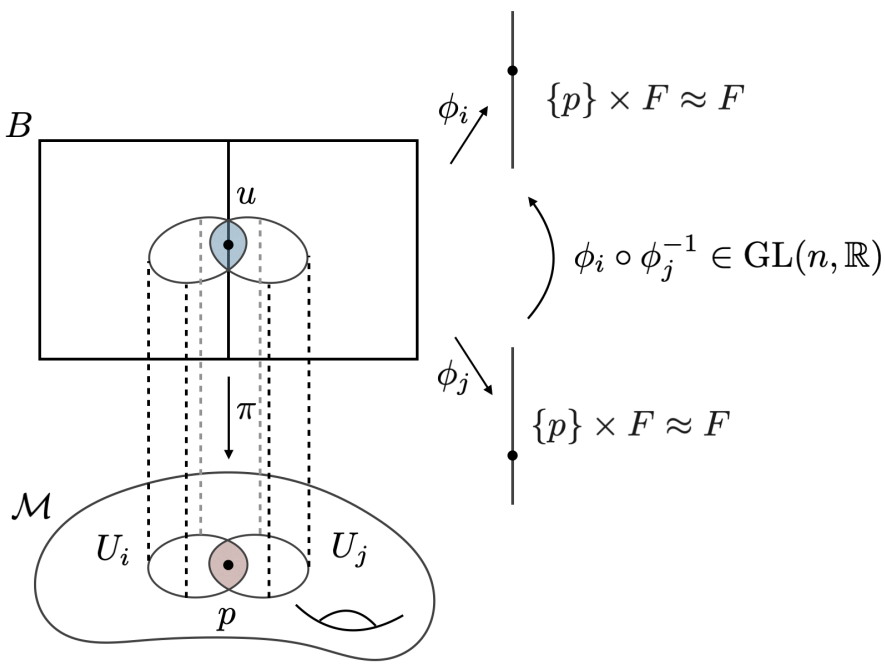


Fig. B.8: A transition function $g_{ij} \coloneqq \phi_i(u) \circ \phi_j^{-1}(u) : U_i \cap U_j \longrightarrow G$.

We note that for a *vector bundle* and a *principal fibre bundle*, we always prefer to use the specific notation of E and P instead of B, respectively.

Furthermore, in a neighborhood U_i, the local trivilization is not unique. Therefore the element in F is also not unique. The transformation of $f_i \in F$, through $g_i \in G$ over U_i, can be defined by

$$g_i(p) : F \longrightarrow F, \tag{B.109a}$$

$$f_i \longmapsto f_i' = g_i(p) \cdot f_i. \tag{B.109b}$$

Because the transformation is discussed in a particular coordinate chart which implies that the coordinate $\{x^\mu(p)\}$ of the point p is fixed, then such a transformation is called *internal* in physics.

Example. (U(1) gauge transformation)

Classically, given an *electrical scalar potential* $\phi(t, \vec{x})$ and a *magnetic vector potential* $\vec{A}(t, \vec{x})$ in the space of $A^1(E)$ over U_i, where E is a vector bundle of U(1) group. we have other physical choices of ϕ and $\vec{A}$ through the transformations

$$\phi \longrightarrow \phi' = \phi + \partial\theta/\partial t \tag{B.110a}$$

and

$$\vec{A} \longrightarrow \vec{A}' = \vec{A} + \vec{\nabla}\theta, \tag{B.110b}$$

respectively, where the function $\theta = \theta(t, \vec{x})$ is so-called gauge function. This implies that there exist an arbitrary function $\theta(x_i)$ such that $\phi(x_i)$ and $\vec{A}(x_i)$ transform to $\phi'(x_i)$ and $\vec{A}'(x_i)$ over a particular coordinate chart (U_i, φ_i) respectively. In U(1) theory, $f_i = \big(\phi(x_i), \vec{A}(x_i)\big)$ is on a vector space of fibre $F = A^1(\mathfrak{u}(1))$. It would lead to the transformation

$$f_i \longmapsto f_i' = \mathrm{Ad}_{g_i(p)}(f_i) + g_i(p)\, dg_i^{-1}(p). \tag{B.111}$$

The U(1) bundel is a vector bundle $E = P \times_{\mathrm{Ad}} \mathfrak{u}(1) = P \times_{\mathrm{Ad}} \mathfrak{u}(1)/\,\mathrm{U}(1)$. It will be clear when we introduce the *associated vector bundle* $E = P \times_\rho F$ of a principal bundle P.

Given another neighborhood U_j, there exists a map $g_j(p)$ such that $f_j' = g_j(p) \cdot f_j$. By using the transition function, we can also have $f_i' = g_{ij}'(p) \cdot f_j'$ through a new transition function $g_{ij}'(p)$. By equaling two relations of f_i', we obtain

$$f_i' = g_{ij}'(p) \cdot f_j' = g_{ij}'(p) \cdot g_j(p) \cdot f_j \tag{B.112a}$$

$$= g_i(p) \cdot f_i = g_i(p) \cdot g_{ij}(p) \cdot f_j. \tag{B.112b}$$

Consequently $g_i(p)$ and $g_j(p)$ over U_i and U_j respectively induce a *transformation* by general transition functions from U_j to U_i

$$\boxed{g_{ij}'(p) = g_i(p) \cdot g_{ij}(p) \cdot g_j^{-1}(p).} \tag{B.113}$$

Example. (Galilean bundle)

A bundle B of the space of the Galilean physics can be realized as the product of the base space, the time $T = \mathbb{R}$ and a fibre, the spatial space $\mathbb{R}^3$.

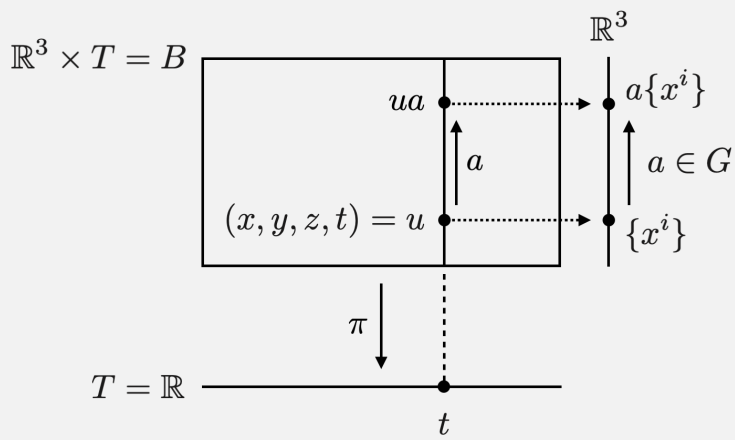


Fig. B.9: A Galilean bundle.

Cross section. A *cross section* or simply called a *section* is a map from $\mathcal{M}$ to B:

$$s : \mathcal{M} \longrightarrow B \tag{B.114}$$

such that $\pi \circ s(p) = p \in \mathcal{M}$, i.e., $\pi \circ s = \mathrm{id}_{\mathcal{M}}$. However, the bundle B in general is not globally trivial. In stead, we restrict $\mathcal{M}$ to a local neighborhood U_i. For $p \in U_i$, we define a *local section*

$$s_i : U_i \longrightarrow \pi^{-1}(U_i), \tag{B.115a}$$

$$p \longmapsto s_i(p), \tag{B.115b}$$

such that $\pi \circ s_i(p) = p$, i.e., $\pi \circ s_i = \mathrm{id}_{U_i}$. Particularly, the section is called a *zero* section if $s_i(p) = 0 \in \pi^{-1}(p)$.

Tensor bundle. Let's discuss the tensor bundle first. Suppose $\mathcal{M}$ is a smooth n-dimensional manifold, the tangent and cotangent spaces of $\mathcal{M}$ at p are $T_p\mathcal{M}$ and $T_p^*\mathcal{M}$. The (r, s)-type tensor space is

$$\mathfrak{T}^r_{s,p} = \underbrace{T_p\mathcal{M} \otimes \cdots \otimes T_p\mathcal{M}}_{r \text{ terms}} \otimes \underbrace{T_p^*\mathcal{M} \otimes \cdots \otimes T_p^*\mathcal{M}}_{s \text{ terms}}. \tag{B.116}$$

We can collect all the tensor space $\mathfrak{T}^r_{s,p}$ over all points $p \in \mathcal{M}$, which is called a (r, s)-type tensor bundle on $\mathcal{M}$

$$\mathfrak{T}^r_s = \bigcup_{p \in \mathcal{M}} \mathfrak{T}^r_{s,p}. \tag{B.117}$$

Let $B = \mathfrak{T}^r_s$ and $F = V^r_s$ the space of all (r, s)-type tensor. Consider the coordinate neighborhood U on $\mathcal{M}$ with the local coordinates $\{x^\mu\}$, a tensor can be written as

$$K = K^{\mu_1 \cdots \mu_r}{}_{\nu_1 \cdots \nu_s} e_{\mu_1} \otimes \cdots \otimes e_{\mu_r} \otimes \theta^{\nu_1} \otimes \cdots \otimes \theta^{\nu_s}, \tag{B.118}$$

The tensor is consist of

- a *total space* (or a *bundle space*) $B = \mathfrak{T}^r_s$;

- a *base space* $\mathcal{M}$;
- a *bundle projection* $\pi(T) = p$ for $T \in B$;
- a *fibre* $B_p = \pi^{-1}(p)$ over p is identified as $\mathfrak{T}^r_{s,p}$ over p;
- a *transition function* $g_{ij} = \phi_i \circ \phi_j^{-1} \in G = \mathrm{GL}(V^r_s)$, which acts on $F = V^r_s$ on the left;

Given a family of coordinate neighborhoods $\{U_i\}$, the tensor in a local coordinates is

$$K_i{}^{\mu'_1\cdots\mu'_r}{}_{\nu'_1\cdots\nu'_s} = \underbrace{(g_{ij})^{\mu'_1}{}_{\mu_1}\cdots(g_{ij})^{\mu'_r}{}_{\mu_r}}_{r\text{ terms}}\underbrace{(g_{ij})_{\nu'_1}{}^{\nu_1}\cdots(g_{ij})_{\nu'_s}{}^{\nu_s}}_{s\text{ terms}} K_j{}^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s} \tag{B.119}$$

There are two special tensor bundles. Let $r = 1$ and $s = 0$, we obtain a *tangent bundle* from a tensor bundle over $\mathcal{M}$

$$T\mathcal{M} = \mathfrak{T}^1_0 = \bigcup_{p\in\mathcal{M}} T_p\mathcal{M}, \tag{B.120}$$

while for $r = 0$ and $s = 1$, we can obtain a *cotangent bundle* over $\mathcal{M}$

$$T^*\mathcal{M} = \mathfrak{T}^0_1 = \bigcup_{p\in\mathcal{M}} T^*_p\mathcal{M}\,. \tag{B.121}$$

Vector bundle. We will focus on the special *tensor bundle*, the *tangent bundle*, which is a *vector bundle* usually denoted by a triple $(E, \mathcal{M}, \pi)$

- a *total space* $E = T\mathcal{M}$;
- a *base space* $\mathcal{M}$;
- a *bundle projection* $\pi(v) = p$ for $v \in E$;
- a *fibre* $E_p = \pi^{-1}(p)$ over $p \in U$ is identified as $T_p\mathcal{M}$ over p;
- a *transition function* $g_{ij} = \phi_i \circ \phi_j^{-1} \in \mathrm{GL}(n;\mathbb{R})$, which acts on $F = \mathbb{R}^n$ on the left.

We note that the fibre $E_p = \pi^{-1}(p)$ over p is isomorphic to the *standard fibre* (or *typical fibre*) F. Consider a family of neighborhood $\{U_i\}$ at $p \in \mathcal{M}$, by local trivializations, we have

$$v = \frac{\partial}{\partial x_i^\mu}\, v_i^\mu = \frac{\partial}{\partial x_j^\nu}\, v_j^\nu \in E_p = T_p\mathcal{M} \tag{B.122}$$

such that $\phi_i(v) = \big(\pi(v), \psi_i(v)\big) = (p, f_i)$ and $\phi_j(v) = \big(\pi(v), \psi_j(v)\big) = (p, f_j)$ over U_i and U_j, respectively, where

$$f_i = \begin{pmatrix} v_i^1 \\ v_i^2 \\ \vdots \\ v_i^n \end{pmatrix} \equiv \mathsf{v}_i \quad \text{and} \quad f_j = \begin{pmatrix} v_j^1 \\ v_j^2 \\ \vdots \\ v_j^n \end{pmatrix} \equiv \mathsf{v}_j. \tag{B.123}$$

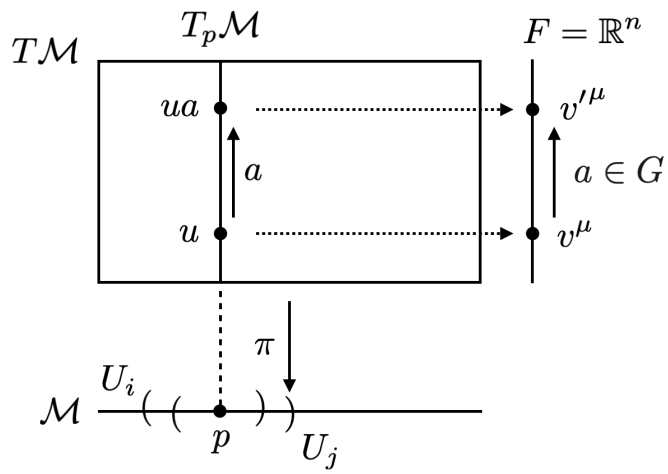


Fig. B.10: A tangent bundle $T\mathcal{M}$.

by local trivializations, the coordinate transformation on F is given by

$$f_i = g_{ij} f_j \quad \text{or} \quad v_i^\mu = (g_{ij})^\mu{}_\nu \, v_j^\nu, \tag{B.124}$$

where the transformation matrix is

$$(g_{ij})^\mu{}_\nu = \frac{\partial x_i^\mu}{\partial x_j^\nu} \equiv \begin{pmatrix} \frac{\partial x_i^1}{\partial x_j^1} & \cdots & \frac{\partial x_i^1}{\partial x_j^n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_i^n}{\partial x_j^1} & \cdots & \frac{\partial x_i^n}{\partial x_j^n} \end{pmatrix} := \boldsymbol{g}_{ij} \tag{B.125}$$

by realizing $g_{ij} \in \mathrm{GL}(n;\mathbb{R})$ acts on F from the *left*, i.e., a matrix transformation of $\mathsf{v}_j \longmapsto \mathsf{v}_i = \boldsymbol{g}_{ij}\mathsf{v}_j$. On the other hand, the transformation of F induces a transformation of the natural frame

$$\frac{\partial}{\partial x_i^\mu} = \frac{\partial}{\partial x_j^\nu} (g_{ij}^{-1})^\nu{}_\mu = \frac{\partial}{\partial x_j^\nu} (g_{ji})^\nu{}_\mu, \quad \text{i.e.,} \quad \mathsf{e}_i = \mathsf{e}_j \, \boldsymbol{g}_{ij}^{-1} = \mathsf{e}_j \, \boldsymbol{g}_{ji}, \tag{B.126}$$

which implies that G acts on a frame from the *right*.

Principal fibre bundle. Let $F = G$ as a typical fibre. A fibre bundle denoted by $P(\mathcal{M}, G, \pi)$ or $P(\mathcal{M}, G)$ (or simply P) is called a *principal fibre bundle* or a G-bundle. We define the group G acting on P on the *right*:

$$R_a : P \longrightarrow P, \tag{B.127}$$

$$u \longmapsto ua, \tag{B.128}$$

and

$$P \times G \longrightarrow P, \tag{B.129}$$

$$(u, a) \longmapsto ua. \tag{B.130}$$

Thus this gives the orbit of G through u: $uG = \{ug \mid g \in G\}$.

For each point $u \in \pi^{-1}(U)$ with $U \subset \mathcal{M}$, the local trivialization is

$$\phi : \pi^{-1}(U) \longrightarrow U \times G, \tag{B.131}$$

$$u \longmapsto \phi(u) = \big(\pi(u), \psi(u)\big) = (p, a), \tag{B.132}$$

and $\phi(ua) = \phi(u)a$ with $a \in G$ is under a local trivialization such that $\pi(u) = \pi(ua) = p$. We note that a point $u = s \cdot a$ can be determined by a unique a acts on a particular local section $s = s(p)$, therefore, G is isomorphic to $\pi^{-1}(p)$. The local trivialization of u is

$$\phi(u) = \phi(sa) = (p, a). \tag{B.133}$$

Hence, by choosing $a = e$ the identity element, we obtain $u = s$. The local trivialization of a section

$$\phi(s) = \phi(se) = (p, e) \tag{B.134}$$

is called the *canonical local trivialization* of P. Let $P = \mathcal{M} \times G$, then R_b for $b \in G$ maps $(p, a) \in \mathcal{M} \times G$ into $(p, ab) \in \mathcal{M} \times G$. Thus we obtain a *trivial* bundle $P(\mathcal{M}, G)$.

Associated fibre bundle. Given another fibre F on which the structure group G of P acts as usual on the *left*

$$G \times F \longrightarrow F, \tag{B.135}$$

$$(g, f) \longmapsto \rho_{g^{-1}} f, \tag{B.136}$$

where $\rho : G \longrightarrow GL(F)$ is a representation (resp. realization) of the structure group G We can construct a fibre bundle $B(\mathcal{M}, F, G, P)$ associated with the principal bundle P, an *associated fibre bundle* over $\mathcal{M}$.

Let G acts on the *right* of the product manifold $P \times F$. The quotient space of $P \times F$ by the equivalence relation induced by G is denoted by $B = (P \times F)/G := P \times_\rho F$, i.e.,

$$g : (u, f) \longmapsto (u, f)g = (ug, \rho_{g^{-1}} f) \sim (u, f) \quad \text{for } g \in G. \tag{B.137}$$

An element of such a equivalence class to (u, f) can be denoted by $[(u, f)]$ in $P \times_\rho F$. Then we define a projection $\pi_B : B \longrightarrow \mathcal{M}$ such that $\pi_B([(u, f)]) = p = \pi(u)$. So $\pi_B^{-1}(p)$ for $p \in \mathcal{M}$ is called the fibre of B over $\mathcal{M}$. Let G act on $\pi^{-1}(U) \times F$ on the right, by local trivilization $\pi^{-1}(U) \approx U \times G$, we have a map

$$(p, a, f) \longmapsto (p, ab, \rho_{b^{-1}} f) \quad \text{for} \quad (p, a, f) \in U \times G \times F \quad \text{and} \quad b \in G. \tag{B.138}$$

So $\pi^{-1}(U) \approx U \times G$ induces an isomorphism $\pi_B^{-1}(U) \approx U \times F$. If $\pi_B^{-1}(U)$ is also diffeomorphic to $U \times F$, then π_E is a differentiable mapping of B onto $\mathcal{M}$. Therefore, we call $B(\mathcal{M}, F, G, P)$ the *fibre bundle associated with the principal fibre bundle* P.

Particularly, if F is a k-dimensional *vector space* $\mathbb{R}^k$, then we call $B = E(\mathcal{M}, F, G, P)$ the *associated vector bundle* of P.

B.7 Connection and Covariant Derivative

Let $P(\mathcal{M}, G)$ be a principal fibre bundle over a manifold $\mathcal{M}$. Given a point $u \in P$, we have a tangent space $T_u P$.

Connection (*Ehresmann* Connection). A connection Γ is an assignment of a subspace H_u of T_uP such that:

(i) $T_uP = V_uP + H_uP$;

(ii) $H_{ua}P = (R_a)_* H_uP$ is induced by $R_a u = ua$ for $a \in G$, which implies that H_uP is invariant under the action of G;

(iii) $H_{ua}P$ depends differentiably on u.

Here V_uP and H_uP are called *vertical* and *horizontal* subspace of T_uP respectively. Consider a 1-parameter group of G, an orbit of a point $\tilde{\gamma}_u(t) = u \in P$ is obtained by ua_t through $a_t = \exp tA \in G$ with $A \in \mathfrak{g}$. The tangent vector of the orbit in V_uP is then reads as

$$A_u^* = \frac{d}{dt}(u \exp tA)|_{t=0}, \tag{B.139}$$

which is called a *fundamental vector field* on P. It is obvious that A^* and A are one-to-one correspondence and they both lie in V_uP. Therefore, given a connection Γ in P, one can define a $\mathfrak{g}$-valued 1-form ω on P which would sends a vector field X into $\mathfrak{g}$

$$\omega : V_uP \longrightarrow \mathfrak{g}, \tag{B.140a}$$

$$A^* \longmapsto \omega(A^*) = A. \tag{B.140b}$$

On the other hand, if $\omega(X) = 0$, it implies that X is horizontal. Then the horizontal subspace can be defined by the kernal of ω, i.e.,

$$H_uP = \{X \in T_uP \mid \omega(X) = 0\}. \tag{B.141}$$

Therefore, for any vector field, we have a decomposition

$$X = vX + hX, \tag{B.142}$$

with two projection maps

$$v : T_uP \longrightarrow V_uP, \tag{B.143a}$$

$$h : T_uP \longrightarrow H_uP. \tag{B.143b}$$

With the identification of a fundamental vector field

$$vX = A^*, \tag{B.144}$$

we have $\omega(vX) = \omega(A^*) = A \in \mathfrak{g}$ and $\omega(hX) = 0$.

Affine connections.

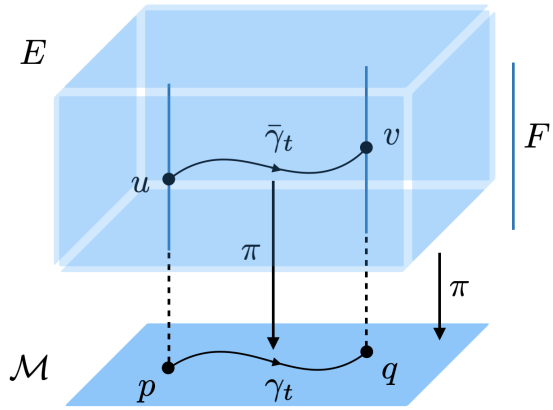


Fig. B.11: The covariant derivative of a section on a vector bundle E.

Covariant derivative. Let $\gamma(t) \equiv \gamma_t$ with $a < t < b$, be a curve, such that $\gamma_a = p$ and $\gamma_b = q$. Define τ_t^{t+h} a *parallel displacement along the curve* γ_t from fibre $\pi^{-1}(\gamma_{t+h})$ to $\pi^{-1}(\gamma_t)$ through the horizontal lift $\gamma^* = \bar{\gamma}_t$ in a vector bundle E, i.e.,

$$\tau_t^{t+h} : \pi^{-1}(\gamma_{t+h}) \longrightarrow \pi^{-1}(\gamma_t). \tag{B.145}$$

Given a section $s(\gamma_{t+h})$ of B, we have a *parallel transported section*

$$s_{\parallel}(\gamma_t) \coloneqq \tau_t^{t+h}\, s(\gamma_{t+h}). \tag{B.146}$$

Therefore the covariant derivative of s in the direction of $\dot{\gamma}_t$ can be defined by

$$\boxed{\nabla_{\dot{\gamma}_t} s = \lim_{h\to 0} \frac{1}{h}\left[\tau_t^{t+h}\, s(\gamma_{t+h}) - s(\gamma_t)\right]} = \frac{\boldsymbol{d}s_{\parallel}(\gamma_t)}{dt} \in \pi^{-1}(\gamma_t). \tag{B.147}$$

Connection. Let E be a vector bundle over $\mathcal{M}$. We consider a vector $X = e_\mu X^\mu$ with $e_\mu = \partial_\mu$ in the natural basis, which gives the differential of a vector written by $\boldsymbol{d}X$.[†17] The differential operator $\boldsymbol{d}$ is a one-form *endomorphism* of $\mathfrak{X}(\mathcal{M})$, denoted by $\mathrm{End}(\mathfrak{X}(\mathcal{M}))$ (or $A^1(\mathfrak{X}(\mathcal{M}))$). Hence, the differential of a basis vector $\boldsymbol{d}e_\mu$ is also a vector in $\mathfrak{X}(\mathcal{M})$. More precisely, it is a vector valued 1-form, which can be expanded by the basis $\{e_\mu\}$ with expanding coefficient 1-form $\{\Gamma^\nu{}_\mu\}$, written by

$$\boxed{\boldsymbol{d}e_\mu = e_\nu\, \Gamma^\nu{}_\mu.} \tag{B.148}$$

The expansion coefficients

$$\boxed{\Gamma^\mu{}_\nu = \Gamma^\mu{}_{\nu\rho}\, dx^\rho} \tag{B.149}$$

[†17] The differential of a vector $X \in \Gamma(E)$ is always denoted by DX or ∇X in most textbooks of differential geometry instead of $\boldsymbol{d}X$. Here we follow the notation of Flander, the differential operator $\boldsymbol{d}$ is also the notation used by Cartan, where $\boldsymbol{d} : A^0(E) \coloneqq \mathfrak{F}(\mathcal{M}) \otimes \Gamma(E) \longrightarrow A^1(E) \coloneqq A^1(\mathcal{M}) \otimes \Gamma(E)$. One can generalize the action of this operator to degree of r, i.e., $\boldsymbol{d} : A^r(E) \coloneqq A^r(\mathcal{M}) \otimes \Gamma(E) \longrightarrow A^{r+1}(E) \coloneqq A^{r+1}(\mathcal{M}) \otimes \Gamma(E)$. The differential operator $\boldsymbol{d}$ is called the *covariant exterior differentiation*.

are called the *connection* 1-*forms* and $\Gamma^\mu{}_{\nu\rho}$ are the corresponding *connection coefficients*. It leads the vector $\boldsymbol{d}X$ to be

$$\begin{aligned}\boldsymbol{d}X &= (\boldsymbol{d}e_\mu)X^\mu + e_\mu(\boldsymbol{d}X^\mu)\\ &= (e_\nu\,\Gamma^\nu{}_\mu)X^\mu + e_\mu(dX^\mu)\\ &= e_\mu(dX^\mu + \Gamma^\mu{}_\nu X^\nu).\end{aligned} \tag{B.150}$$

However, the differential generates a 1-form, the vector $\boldsymbol{d}X$ can be expressed as

$$\boldsymbol{d}X = e_\mu(\partial_\rho X^\mu + \Gamma^\mu{}_{\nu\rho}X^\nu)dx^\rho \equiv (\boldsymbol{d}X)^\mu{}_\rho\, e_\mu \otimes dx^\rho. \tag{B.151}$$

The partial derivative is always denoted by a subscript of *comma* as

$$\partial_\rho X^\mu \equiv X^\mu{}_{,\rho}, \tag{B.152}$$

so we write the component $(\boldsymbol{d}X)^\mu{}_\rho$ as

$$\boxed{(\boldsymbol{d}X)^\mu{}_\rho \coloneqq X^\mu{}_{;\rho} \equiv \nabla_\rho X^\mu = X^\mu{}_{,\rho} + \Gamma^\mu{}_{\nu\rho}X^\nu,} \tag{B.153}$$

which is called the *covariant derivative* of a vector component X^μ in the direction of e_ρ and represented by the subscript of *semicolon*. As a result, the differential operator $\boldsymbol{d}$ is called the *connection* of a *vector* and always denoted by ∇ if $\boldsymbol{d}X$ is a tangent vector on $\mathcal{M}$, i.e.,

$$\nabla X \equiv \boldsymbol{d}X \quad \text{if} \quad \boldsymbol{d}X \in T\mathcal{M}. \tag{B.154}$$

Gauss's formula. If X is a vector on the smooth submanifold $\mathcal{M} \subset \overline{\mathcal{M}}$ with dimension $\dim \mathcal{M} = m$ the differential dX, in general, would contain components in the normal space $\mathcal{M}^\perp \subset \overline{\mathcal{M}}$ with the codimension $\dim \mathcal{M}^\perp = n$, then we have the decomposition given by the so-called *Gauss's formula*

$$\boldsymbol{d}X = (\boldsymbol{d}X)_t + (\boldsymbol{d}X)_n \equiv \nabla X + A(X), \tag{B.155}$$

where A is a vector-valued 1-form called the *shape operator* of the submanifold $\mathcal{M}$ defined by a set of *unit normal* vector $\{N_\alpha\}$ for $\alpha = 1, \ldots, n$. Hence, A_X should be defined by

$$A(X) \coloneqq N_\alpha S^\alpha(X), \tag{B.156}$$

where $S^\alpha(X)$ is a 1-form, which can be obtained by doing the inner product of $\boldsymbol{d}X$ and N_β:

$$g(\boldsymbol{d}X, N_\beta) = g\big(A(X), N_\beta\big) = g\big(N_\alpha S^\alpha(X), N_\beta\big) = S^\alpha(X)g_{\alpha\beta} = S_\beta(X). \tag{B.157}$$

Apply (B.155) on a vector $Y \in \mathfrak{X}(\mathcal{M})$, the covariant derivative of X is

$$\overline{\nabla}_Y X = \nabla_Y X + A_Y(X), \tag{B.158}$$

where

$$A_Y(X) = N_\alpha S^\alpha(X)(Y) \quad \text{and} \quad S_\beta(X)(Y) = g(\overline{\nabla}_Y X, N_\beta) = -g(X, \overline{\nabla}_Y N_\beta). \tag{B.159}$$

Hence, we can define a $(0,2)$-tensor or a *quadratic form* on $\mathcal{M}$

$$K^\alpha = K^\alpha{}_{\mu\nu}\, dx^\mu \otimes dx^\nu \in V^0_2(\mathcal{M}) \tag{B.160}$$

called the *extrinsic curvature*, which is associated with $A_X(Y)$ and the operator S^α by

$$A_Y(X) = N_\alpha K^\alpha(X,Y) \quad \text{and} \quad K^\alpha(X,Y) \coloneqq S^\alpha(X)(Y). \tag{B.161}$$

Weingarten's formula. Similarly we can consider the differential of N_α, which is assumed to be

$$\boldsymbol{d}N_\alpha = (\boldsymbol{d}N_\alpha)_t + (\boldsymbol{d}N_\alpha)_n \equiv B_{N_\alpha} + \nabla^\perp N_\alpha, \tag{B.162}$$

In addition we have $g(X, N_\beta) = 0$, it leads to $g(\boldsymbol{d}X, N_\beta) = dg(X, N_\beta) - g(X, \boldsymbol{d}N_\beta)$. Hence, we obtain

$$S_\beta(X) = dg(X, N_\beta) - g(X, \boldsymbol{d}N_\beta) = dg(X, N_\beta) - g(X, \nabla N_\beta) \tag{B.163a}$$

or

$$K_\beta(X, Y) = S_\beta(X)(Y) = dg(X, N_\beta) - g(X, \nabla_Y N_\beta). \tag{B.163b}$$

For particular case of a hypersurface $\mathcal{M} \subset \overline{\mathcal{M}}$, we only have one unit normal vector denoted by N and

$$\boldsymbol{d}N = (\boldsymbol{d}N)_t + (\boldsymbol{d}N)_n \equiv \nabla N + (\boldsymbol{d}N)_n, \tag{B.164}$$

Due to $g(N, N) = 1$, $g(\boldsymbol{d}N, N) = 0$, so the Weingarten's formula becomes a simpler form:

$$\boldsymbol{d}N = \nabla N \quad \text{or} \quad \overline{\nabla}_Y N = \nabla_Y N. \tag{B.165}$$

$$A(X) \coloneqq g(N, \boldsymbol{d}X) \tag{B.166a}$$

$$= \boldsymbol{d}\underbrace{g(N, X)}_{0} - g(\boldsymbol{d}N, X) \tag{B.166b}$$

The *extrinsic curvature* is define by

$$K = K^\mu{}_\nu\, e_\mu \otimes dx^\nu \coloneqq -\nabla N \tag{B.167}$$

Then

$$\begin{aligned} A(X) &= g(K, X) - g(\boldsymbol{d}N, X) \\ &= -g(\nabla N + A(N), X) = -g(\nabla N, X). \end{aligned} \tag{B.168}$$

where x^α is the coordinate of the normal space $\mathcal{M}^\perp$ and $e_\alpha = \partial_\alpha \equiv \partial/\partial x^\alpha$. Therefore, we have

$$\begin{aligned} A(X) &= -g(\nabla N, X) = -g(\nabla e_\alpha, X) = -g(\Gamma^\rho{}_\alpha\, e_\rho, X) \\ &= -g(\Gamma^\rho{}_{\alpha\nu}\, e_\rho \otimes dx^\nu + \Gamma^\rho{}_{\alpha\beta}\, e_\rho \otimes dx^\beta, X^\mu\, e_\mu) = -\Gamma_{\mu\alpha\nu} X^\mu\, dx^\nu - \Gamma_{\mu\alpha\beta} X^\mu\, dx^\beta. \end{aligned} \tag{B.169}$$

Let $\mathcal{M}$ be a *hypersurface*, the codimension is one and we only have one coordinate $x^\alpha = x^n$ in $\mathcal{M}^\perp$, which leads to

$$A(X) = -\Gamma_{\mu n\nu} X^\mu\, dx^\nu - \Gamma_{\mu n\beta} X^\mu\, dx^\beta \equiv K^\nu{}_\mu\, e_\nu \otimes dx^\mu \quad \text{and} \quad K^\nu{}_\mu \equiv -\Gamma^\nu{}_{n\mu}. \tag{B.170}$$

We note that

$$\begin{aligned} A(X) &= g(N, \boldsymbol{d}X) = g\big(N, \nabla X + A(X)\big) = g(N, A_X) = g(N, X^\mu \omega^\gamma{}_\mu\, e_\gamma) \\ &= g(N, \Gamma^\gamma{}_{\mu\nu} X^\mu\, e_\gamma \otimes dx^\nu + \Gamma^\gamma{}_{\mu\beta} X^\mu\, e_\gamma \otimes dx^\beta) = \Gamma_{n\mu\nu} X^\mu\, dx^\nu + \Gamma_{n\mu\beta} X^\mu\, dx^\beta, \end{aligned} \tag{B.171}$$

where we have $N = e_n$.

Absolute derivative. We can define the *absolute derivative* by using (B.151) with an arbitrary vector $Y = e_\mu Y^\mu$

$$\frac{\boldsymbol{d}Y}{dt} = e_\mu\,(\boldsymbol{d}Y)^\mu{}_\rho\,\frac{dx^\rho}{dt} = e_\mu(X^\rho\nabla_\rho Y^\mu) \overset{\text{or}}{=} e_\mu(X^\rho Y^\mu{}_{;\rho}). \tag{B.172}$$

In addition, we can define

$$\frac{\boldsymbol{d}Y}{dt} := e_\mu\frac{\nabla Y^\mu}{dt}, \tag{B.173}$$

and then the *absolute derivative* for a vector *component* Y^μ is

$$\boxed{\frac{\nabla Y^\mu}{dt} = X^\rho\nabla_\rho Y^\mu \overset{\text{or}}{=} X^\rho Y^\mu{}_{;\rho}.} \tag{B.174}$$

Furthermore, if the vector Y is an tangent vector along a curve $\gamma(t)$, i.e., $Y = \boldsymbol{d}\mathbf{p}/dt = e_\mu X^\mu = X$, its *acceleration* should be $a = e_\mu a^\mu$ obtained by the absolute derivative of X

$$\boxed{a := \frac{\boldsymbol{d}X}{dt}} \quad \text{with} \quad \boxed{a^\mu = X^\rho\nabla_\rho X^\mu \overset{\text{or}}{=} X^\rho X^\mu{}_{;\rho}.} \tag{B.175}$$

Parallel transport. If Y is a parallel vector, it does not change and always parallel to itself. Therefore, we have

$$\boxed{\boldsymbol{d}Y = 0} \quad \text{or} \quad \boxed{\frac{\boldsymbol{d}Y}{dt} = e_\mu\left\{\frac{dY^\mu}{dt} + \Gamma^\mu{}_{\nu\rho}Y^\nu\frac{dx^\rho}{dt}\right\} = 0.} \tag{B.176}$$

and its equation of components should be

$$\boxed{(\boldsymbol{d}Y)^\mu{}_\rho = \nabla_\rho Y^\mu = Y^\mu{}_{;\rho} = Y^\mu{}_{,\rho} + \Gamma^\mu{}_{\nu\rho}Y^\nu = 0} \tag{B.177}$$

Obviously, the parallel transport of the tangent vector $Y = X$ leads to the vanished acceleration

$$\boxed{a^\mu = X^\rho\nabla_\rho X^\mu = X^\rho X^\mu{}_{;\rho} = \frac{\nabla X^\mu}{dt} = 0} \tag{B.178}$$

Parallel propagator. Let E be a vector bundle and $s = e_i s^i$ be the section on E. If s Through the parallel transport of a section s, we have

$$ds^i = \delta s^i = -\omega^i{}_j\,s^j \qquad \Longrightarrow \qquad \frac{ds^i}{d\lambda} = -\omega^i{}_{jk}\,\frac{\vartheta^k}{d\lambda}\,s^j := A^i{}_j(\lambda)\,s^j. \tag{B.179}$$

Then we can write down the section at λ in terms of the integral form

$$s^i(\lambda) = s^i(\lambda_0) - \int_{\lambda_0}^{\lambda} d\lambda'\,A^i{}_j(\lambda')\,s^j(\lambda') \tag{B.180}$$

$$= s^i(\lambda_0) - \int_{\lambda_0}^{\lambda} d\lambda'\,A^i{}_j(\lambda')\left[s^j(\lambda_0) - \int_{\lambda_0}^{\lambda'} d\lambda''\,A^j{}_k(\lambda'')\left[s^k(\lambda_0) - \cdots\right]\right] \tag{B.181}$$

Now define the so-called path ordering operation, by

$$\mathcal{P}^i{}_j(\lambda,\lambda_0) := -\omega^i{}_{jk}\frac{\vartheta^k}{d\lambda} \tag{B.182}$$

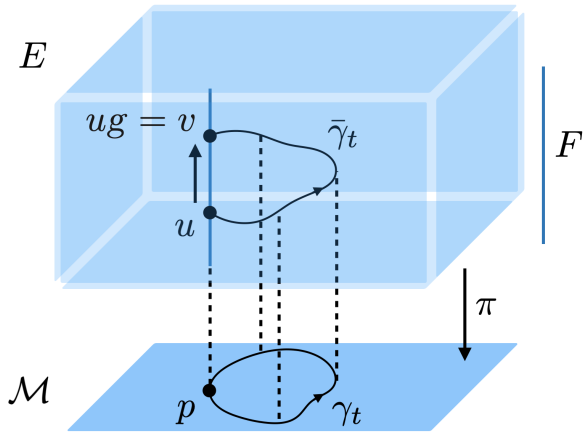


Fig. B.12: Holonomy on a bundle E.

B.8 Holonomy

Given two points $p, q \in \mathcal{M}$ joining by a path γ in the base space $\mathcal{M}$, the corresponding points $u = \pi^{-1}(p) \in B_p$ and $v = \pi^{-1}(q) \in B_q$ in the bundle space B are joining by the lift $\gamma^* = \bar{\gamma}_t$ of $\gamma(t)$. If we restrict $q = p \in \mathcal{M}$, the path γ forms a closed loop in $\mathcal{M}$. However, in the the bundle space B, we should have $u = \pi^{-1}(p)$ and $v = \pi^{-1}(p)$ such that $v = ug$ with $g \in \Phi$ and $\pi(u) = p = \pi(ug)$, where Φ is called the *Holonomy group*, which is a subgroup of the structure group G.

B.9 Metric Tensor and Geodesic Equation

Metric tensor. We define a symmetric $(0, 2)$-tensor field (a quadratic differential form) called metric tensor on a manifold $\mathcal{M}$

$$g = g_{\mu\nu}\, dx^\mu \otimes dx^\nu. \tag{B.183}$$

The component $g_{\mu\nu}$ can be obtained by applying g to two coordinate base vectors

$$g(e_\mu, e_\nu) = g_{\mu\nu}. \tag{B.184}$$

We can find a coordinate system to make the metric tensor diagonal and the eigenvalue is equal to 1 or -1. If all values are $+1$, the manifold $\mathcal{M}$ is called *Riemannian*, otherwise it is called *Lorentzian*. For a Lorentzian manifold the *canonical* form of the metric tensor would contain value -1 for s terms and $+1$ for t terms of the canonical form

$$g_{\mu\nu} \overset{*}{=} \operatorname{diag}\big(\underbrace{-1, \ldots, -1}_{s=\frac{1}{2}(n-q)\text{ terms}}, \overbrace{+1, \ldots, +1}^{t=\frac{1}{2}(n+q)\text{ terms}}\big), \tag{B.185}$$

where $*$ means the particular coordinates is chosen. The number $n = s+t$ should be the dimensions of $\mathcal{M}$ and $q = -s+t$ being the trace of the metric tensor is also called the signature of the metric. Particularly, for $s = 1$, the manifold is called *pseudo-Riemannian* or *semi-Riemannian*. We define

a canonical form by $\vartheta = e_\mu dx^\mu \in \mathrm{End}(T\mathcal{M})$ which is an identity map for every vector $X \in \mathfrak{X}(\mathcal{M})$, i.e.,

$$\vartheta(X) = e_\mu \underbrace{dx^\mu(e_\nu X^\nu)}_{\delta^\mu_\nu X^\nu = X^\mu} = X. \tag{B.186}$$

Physically we write the line element to represent a metric tensor g as a inner product between two canonical forms

$$ds^2 = g(\vartheta, \vartheta) = g_{\mu\nu}\, dx^\mu dx^\nu. \tag{B.187}$$

Geodesic equations. The geodesic equation can be obtained by extrimizing the arc length between points p and q

$$L = \int_p^q ds = \int_p^q \sqrt{g_{\mu\nu}\, dx^\mu dx^\nu} = \int_{t_p}^{t_q} dt \sqrt{g_{\mu\nu} \frac{dx^\mu}{dt}\frac{dx^\nu}{dt}}, \tag{B.188}$$

where t is the parameter describing the evolution of a curve $\gamma(t)$. Through the variation, the extreme behavior of the curve gives equation of the so-called *geodesic equations*

$$\frac{d^2x^\rho}{dt^2} + \left\{ {\rho \atop \mu\nu} \right\} \frac{dx^\mu}{dt}\frac{dx^\nu}{dt} = 0, \tag{B.189}$$

where the coefficients in the second term are called *Christoffel symbols*

$$\left\{ {\rho \atop \mu\nu} \right\} = \frac{1}{2} g^{\rho\sigma} \left(\frac{\partial g_{\mu\sigma}}{\partial x^\rho} + \frac{\partial g_{\nu\sigma}}{\partial x^\mu} - \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \right). \tag{B.190}$$

It is obvious that Christoffel symbols are symmetric in μ and ν

$$\left\{ {\rho \atop \nu\mu} \right\} = + \left\{ {\rho \atop \mu\nu} \right\}. \tag{B.191}$$

B.10 Transformation Law of a Connection

According to the basis transformation

$$\mathsf{s}_j \longmapsto \mathsf{s}_i = \mathsf{s}_j\, \boldsymbol{g}_{ji} \tag{B.192}$$

from neighborhood U_j to U_i through $\boldsymbol{g}$ in the bundle B, the transformation law of a connection form is induced as follows:

$$\begin{aligned} \boldsymbol{d}\mathsf{s}_i &= \mathsf{s}_i\, \boldsymbol{\omega}_i = \mathsf{s}_j\, \boldsymbol{g}_{ji}\, \boldsymbol{\omega}_i \\ &= \boldsymbol{d}(\mathsf{s}_j\, \boldsymbol{g}_{ji}) = \boldsymbol{d}\mathsf{s}_j\, \boldsymbol{g}_{ji} + \mathsf{s}_j\, d\boldsymbol{g}_{ji} = \mathsf{s}_j\, \boldsymbol{\omega}_j\, \boldsymbol{g}_{ji} + \mathsf{s}_j\, d\boldsymbol{g}_{ji}. \end{aligned} \tag{B.193}$$

Hence, we obtain

$$\boldsymbol{g}_{ji}\, \boldsymbol{\omega}_i = \boldsymbol{\omega}_j\, \boldsymbol{g}_{ji} + d\boldsymbol{g}_{ji} \quad \Longrightarrow \quad \boxed{\boldsymbol{\omega}_i = \boldsymbol{g}_{ji}^{-1}\, \boldsymbol{\omega}_j\, \boldsymbol{g}_{ji} + \boldsymbol{g}_{ji}^{-1}\, d\boldsymbol{g}_{ji}.} \tag{B.194}$$

In a vector bundle E, we have $\boldsymbol{g} \in \mathrm{GL}(4; \mathbb{R})$

$$\mathsf{s}_j = \mathsf{e}_j = (e_a), \quad \boldsymbol{\omega}_j = (\omega^a{}_b) \quad \boldsymbol{g}_{ji} = \big((A^{-1})^a{}_{a'}\big), \tag{B.195}$$

$$\mathsf{s}_i = \mathsf{e}_i = (e_{a'}), \quad \boldsymbol{\omega}_i = (\omega^{a'}{}_{b'}), \quad \boldsymbol{g}_{ji}^{-1} = (A^{a'}{}_a), \tag{B.196}$$

so, we have the transform of the connection form

$$\boxed{\omega^{a'}{}_{b'} = A^{a'}{}_a\, \omega^a{}_b\, (A^{-1})^b{}_{b'} + A^{a'}{}_c\, d(A^{-1})^c{}_{b'}.} \tag{B.197}$$

B.11 Summary of Torsion and Curvature Tensors

By transforming from one local coordinates to another, the transformation law of a connection form $\Gamma^{\rho}{}_{\mu} = \Gamma^{\rho}{}_{\mu\nu}\, dx^{\nu}$ in coordinate frame becomes

$$\begin{aligned}\underbrace{\Gamma^{\rho'}{}_{\mu'}}_{\Gamma^{\rho'}{}_{\mu'\nu'}\, dx^{\nu'}} &= A^{\rho'}{}_{\rho}\, \Gamma^{\rho}{}_{\mu}\, (A^{-1})^{\mu}{}_{\mu'} + A^{\rho'}{}_{\sigma}\, d(A^{-1})^{\sigma}{}_{\mu'} \\ &= A^{\rho'}{}_{\rho}\, \Gamma^{\rho}{}_{\mu\nu}\, (A^{-1})^{\mu}{}_{\mu'}\, \underbrace{dx^{\nu}}_{(A^{-1})^{\nu}{}_{\nu'}\, dx^{\nu'}} + A^{\rho'}{}_{\sigma}\, \frac{\partial}{\partial x^{\nu}} (A^{-1})^{\sigma}{}_{\mu'}\, \underbrace{dx^{\nu}}_{(A^{-1})^{\nu}{}_{\nu'}\, dx^{\nu'}},\end{aligned} \tag{B.198}$$

where we do the transformation of $\boldsymbol{g} = \big((A^{-1})^{\rho}{}_{\rho'}\big) \in \mathrm{GL}(4;\mathbb{R})$. So, by comparing the left and right hand sides of (B.198), we obtain the coordinate transformation law for connection coefficient:

$$\boxed{\Gamma^{\rho'}{}_{\mu'\nu'} = A^{\rho'}{}_{\rho}\, \Gamma^{\rho}{}_{\mu\nu}\, (A^{-1})^{\mu}{}_{\mu'}\, (A^{-1})^{\nu}{}_{\nu'} + A^{\rho'}{}_{\sigma}\, \underbrace{\frac{\partial}{\partial x^{\nu}} (A^{-1})^{\sigma}{}_{\mu'}\, (A^{-1})^{\nu}{}_{\nu'}}_{\frac{\partial^2 x^{\sigma}}{\partial x^{\nu'} \partial x^{\mu'}} = x^{\sigma}{}_{,\,\mu'\nu'}}.} \tag{B.199}$$

The transformation appears an *inhomogeneous* term on the right hand side, it implies that the connection coefficient is not a tensor.

Torsion tensor. Given an affine connection in the *coordinate frame*, we can decompose it into symmetric and antisymmetric part

$$\Gamma^{\rho}{}_{\mu\nu} = \Gamma^{\rho}{}_{(\mu\nu)} + \Gamma^{\rho}{}_{[\mu\nu]}. \tag{B.200}$$

It is obvious that the antisymmetric part of the connection transforms as a tensor:

$$\Gamma^{\rho'}{}_{[\mu'\nu']} = A^{\rho'}{}_{\rho}\, \Gamma^{\rho}{}_{[\mu\nu]}\, (A^{-1})^{\mu}{}_{\mu'}\, (A^{-1})^{\nu}{}_{\nu'}. \tag{B.201}$$

We can define such a antisymmetric part of a connection to be a *torsion* tensor:

$$\boxed{T^{\rho}{}_{\nu\mu} = \Gamma^{\rho}{}_{\mu\nu} - \Gamma^{\rho}{}_{\nu\mu}.} \tag{B.202}$$

Riemann curvature. By using connections, the

- Christoffel-Riemann curvature $R^{\rho}{}_{\sigma\mu\nu}$

$$\begin{aligned}R^{\rho}{}_{\sigma\mu\nu} &= \partial_{\mu}\Gamma^{\rho}{}_{\sigma\nu} - \partial_{\nu}\Gamma^{\rho}{}_{\sigma\mu} + \Gamma^{\rho}{}_{\lambda\mu}\Gamma^{\lambda}{}_{\sigma\nu} - \Gamma^{\rho}{}_{\lambda\nu}\Gamma^{\lambda}{}_{\sigma\mu} \\ &\overset{\text{or}}{=} \Gamma^{\rho}{}_{\sigma\nu,\,\mu} - \Gamma^{\rho}{}_{\sigma\mu,\,\nu} + \Gamma^{\lambda}{}_{\sigma\nu}\Gamma^{\rho}{}_{\lambda\mu} - \Gamma^{\lambda}{}_{\sigma\mu}\Gamma^{\rho}{}_{\lambda\nu}.\end{aligned} \tag{B.203}$$

- Ricci curvature $R_{\sigma\nu}$ or $\mathrm{Ric}_{\sigma\nu}$

$$R_{\sigma\nu} = g^{\rho\mu} R_{\rho\sigma\mu\nu}. \tag{B.204}$$

- Scalar curvature R

$$\mathbf{R} = g^{\sigma\nu} R_{\sigma\nu}. \tag{B.205}$$

- The Weyl tensor is the traceless part of the Riemann tensor defined by

$$C^{\rho\sigma}{}_{\mu\nu} = R^{\rho\sigma}{}_{\mu\nu} - 2\delta^{[\rho}{}_{[\mu}R^{\sigma]}{}_{\nu]} + \frac{1}{3}\delta^{[\rho}{}_{[\mu}\delta^{\sigma]}{}_{\nu]}R. \tag{B.206}$$

  Hence, it can be show that the contraction of Weyl tensor vanishes

$$C^{\mu}{}_{\sigma\mu\nu} = g^{\rho\mu}C_{\rho\sigma\mu\nu} = 0. \tag{B.207}$$

  In *vacuum* Einstein equation gives $R_{\mu\nu} = 0$. Hence, we also have vanished scalar curvature $\mathbf{R} = 0$, which implies that, from (B.206), the Weyl tensor should be the Riemann tensor. Therefore, the spacetime of vacuum is determined by the Weyl tensor.

- Einstein tensor $G_{\sigma\nu}$

$$G_{\sigma\nu} \coloneqq R_{\sigma\nu} - \frac{1}{2}g_{\sigma\nu}R. \tag{B.208}$$

There are some important identities

- Biannchi identity

$$\nabla_{[\lambda}R^{\rho}{}_{\sigma|\mu\nu]} = 0 \quad \text{or} \quad R^{\rho}{}_{\sigma[\mu\nu;\,\lambda]} = 0. \tag{B.209}$$

- Contracted Biannchi identity

$$\nabla_{\sigma}G^{\sigma}{}_{\nu} = 0. \tag{B.210}$$

  Through the Einstein equation, this identity is equivalent to the conservation of energy-momentum stress tensor

$$\nabla_{\sigma}T^{\sigma}{}_{\nu} = 0. \tag{B.211}$$

B.12 Abstract Index Notations

The idea given by R. Penrose is *not* to introduce a basis but to use a notation for tensors that mirrors the expressions for their basis components. The *abstract index notation* of a tensor K of type (r, s) will be denoted by

$$K^{a_1\cdots a_r}{}_{b_1\cdots b_s} \equiv K \in V^r_s. \tag{B.212}$$

We note that the Latin indices do *not* really label the basis components, they are used to recognize the type by position of indices and show the form of the component of a tensor. We can write the tensor by keeping the Latin indices for basis as

$$\boxed{K^{a_1\cdots a_r}{}_{b_1\cdots b_s} = K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s}(e_{\mu_1})^{a_1}\otimes\cdots\otimes(e_{\mu_r})^{a_r}\otimes(dx^{\nu_1})_{b_1}\otimes\cdots\otimes(dx^{\nu_s})_{b_s}.} \tag{B.213}$$

So the abstract index notation $K^{a_1\cdots a_r}{}_{b_1\cdots b_s}$ is similar to the component expression of a tensor $K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s}$, but it contains information of basis. Clearly, when one introduce a basis, the type of tensor is apparently known. Therefore, we can *ignore* the Latin indices and transform the abstract notation to the classical notation of a tensor field

$$\boxed{K^{a_1\cdots a_r}{}_{b_1\cdots b_s} \equiv K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s}e_{\mu_1}\otimes\cdots\otimes e_{\mu_r}\otimes dx^{\nu_1}\otimes\cdots\otimes dx^{\nu_s}.} \tag{B.214}$$

Vectors and covectors. For a type $(1,0)$ tensor, a vector, we can recognize its abstract index notation by

$$V^a = V^\mu (e_\mu)^a \equiv V^\mu \, e_\mu. \tag{B.215}$$

Similarly, a type $(0,1)$ tensor should be

$$\omega_a = \omega_\mu (dx^\mu)_a \equiv \omega_\mu \, dx^\mu. \tag{B.216}$$

Metric tensor. The metric tensor is type $(0,2)$. By abstract index notation, it can be written by

$$g_{ab} = g_{\mu\nu} (dx^\mu)_a \otimes (dx^\nu)_b \equiv g_{\mu\nu} \, dx^\mu \otimes dx^\nu. \tag{B.217}$$